

Deep Learning-enabled Prediction of Intrinsically Disordered Protein Regions

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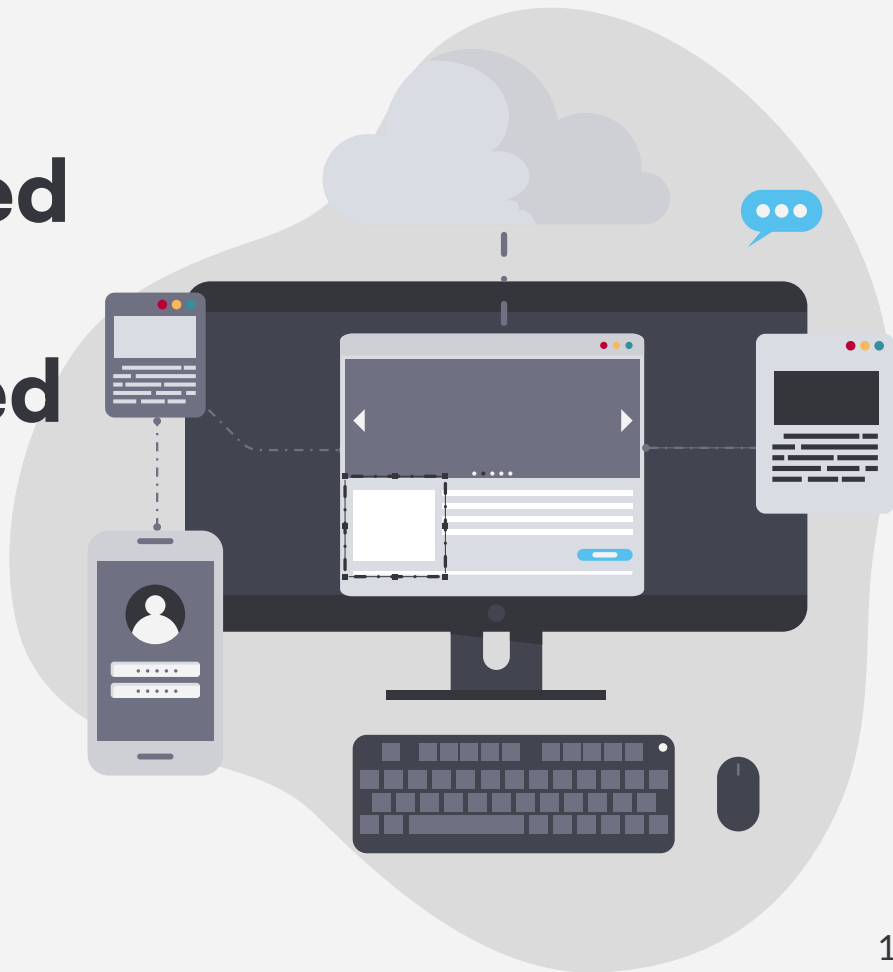




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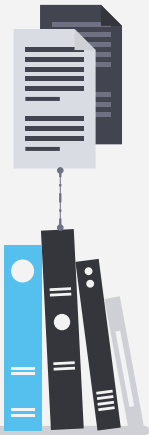
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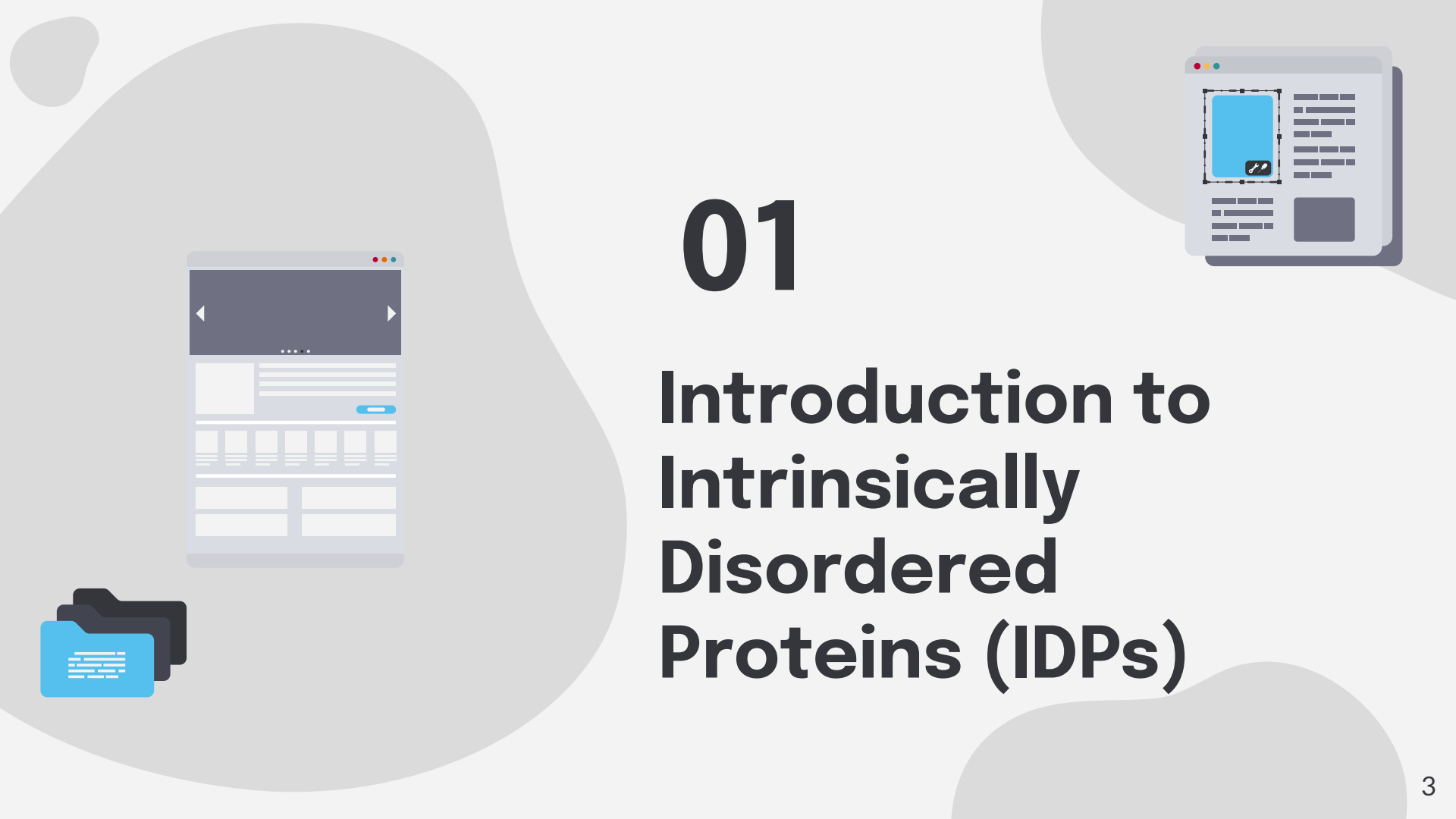
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01

Introduction to Intrinsically Disordered Proteins (IDPs)

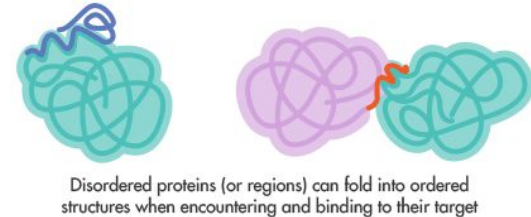
What are IDPs?

Intrinsically disordered proteins lack a fixed, stable 3D structure as opposed to most static proteins we know of



Key roles of IDPs

- Cell signalling and regulation
- Formation of membraneless organelles via Liquid-Liquid phase separation (LLPS)



Key Characteristics of IDPs

- Low complexity
 - Often have repeated amino acid residues
 - Ala, Arg, Gly, Gln, Glu, Lys, Pro, and Ser
- Highly charged or polar amino acid residues
- Low overall hydrophobicity → prevents the folding of hydrophobic core

Liquid-liquid Phase Separation

This process occurs when certain conditions such as pH, concentration or temperature are met. It results in the IDPs transitioning from:



Dynamic Liquid

Functional, reversible droplets. Essential for RNA processing and stress response.



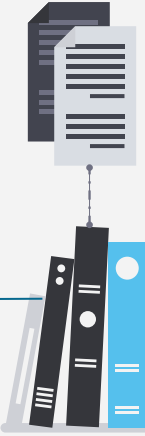
Pathological Hydrogel

Droplets lose fluidity and become semi-solid viscous gels.



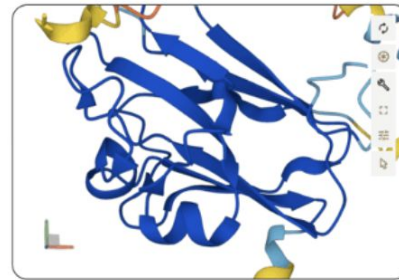
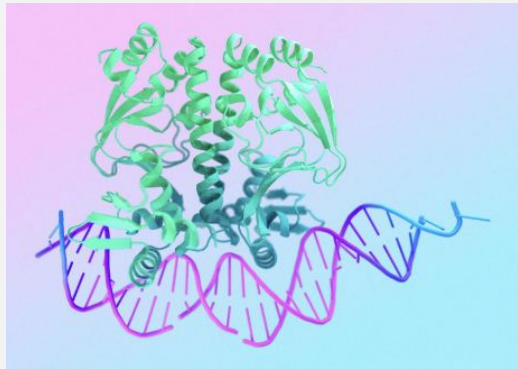
Insoluble Fibrils

These solid aggregates clog cellular machinery, killing neurons.



Machine Learning: Predicting Presence of Intrinsically Disordered Regions

Many existing tools to predict structure or sequences of proteins such as AlphaFold



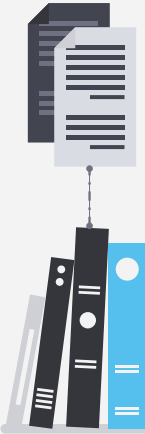
Model Confidence ⓘ

Very high (pLDDT > 90)

Low (70 > pLDDT > 50)

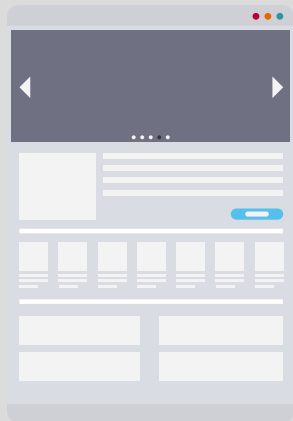
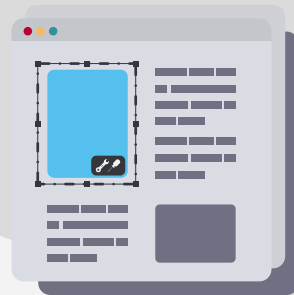
High (90 > pLDDT > 70)

Very low (pLDDT < 50)



02

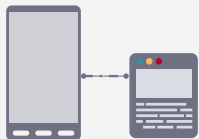
Experimental Methods of IDPs and Limitations





Experimental Characterization of IDPs

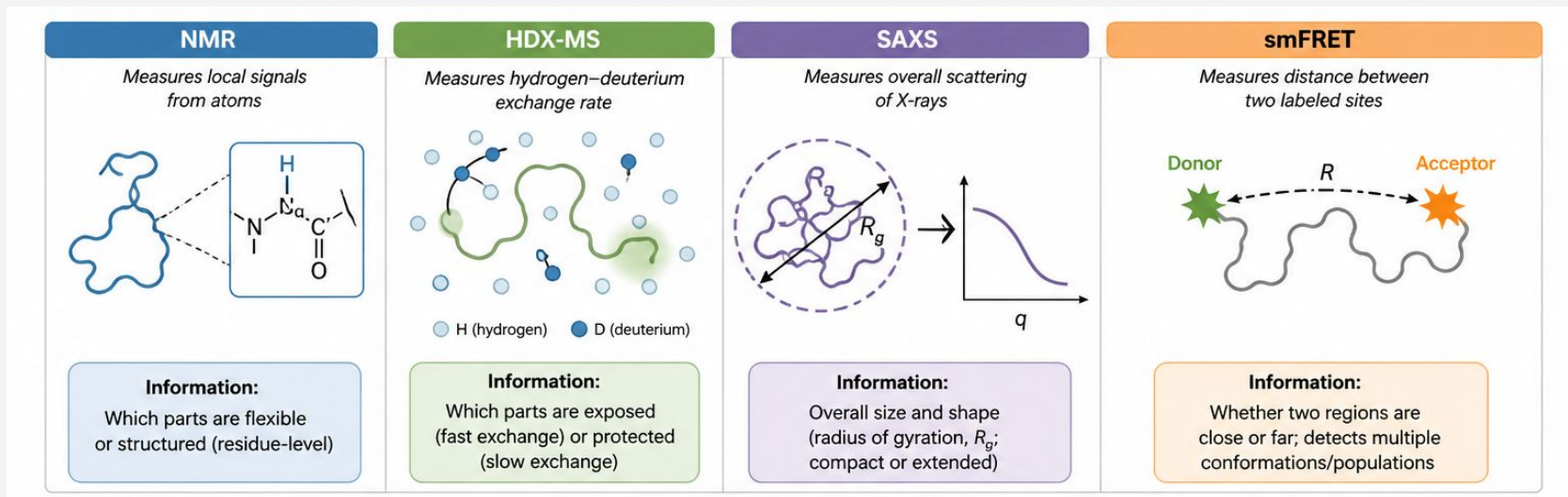
- IDPs lack a fixed 3D structure
- X-ray crystallography is not suitable
- Require solution-based techniques
- Methods:
 - Nuclear Magnetic Resonance (NMR)
 - Hydrogen-Deuterium Exchange Mass Spectrometry (HDX-MS)
 - Small-angle X-ray Scattering (SAXS)
 - Single-molecule Förster Resonance Energy Transfer (smFRET)





Experimental Characterization of IDPs

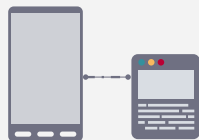
Each method probes different structural properties:





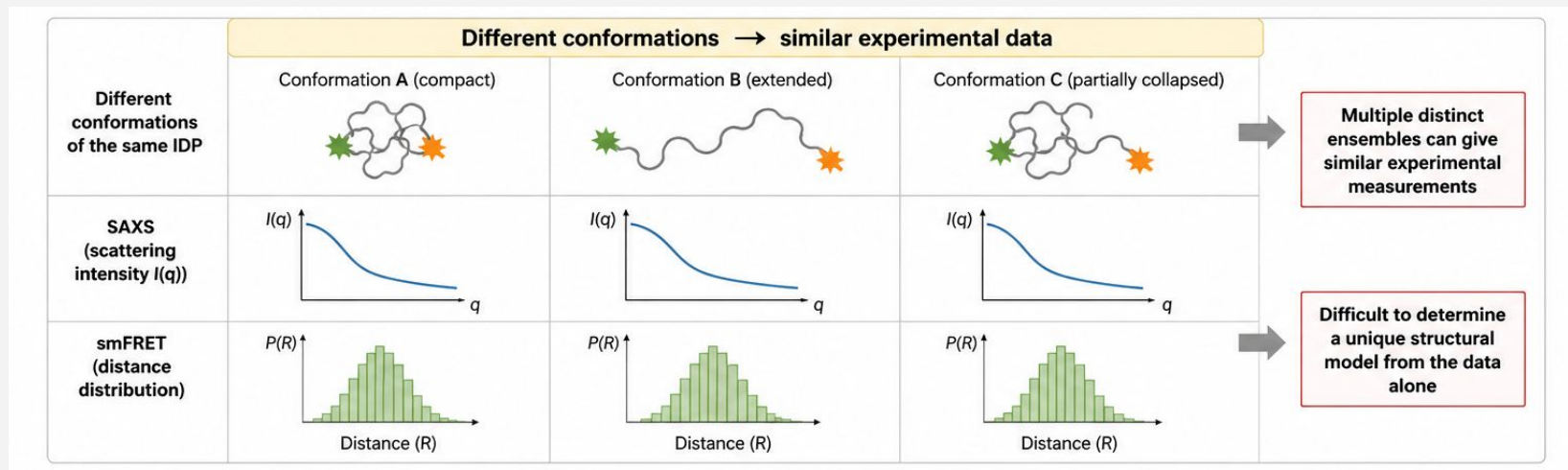
Experimental Characterization of IDPs

Method	What it Measures	What it Tells Us
NMR	Local atomic signals	Which regions are flexible or have transient structure
HDX-MS	Hydrogen exchange rate	Which regions are exposed or protected (flexibility)
SAXS	Overall scattering (R_g)	Overall size and compactness of the protein
smFRET	Distance between labelled sites	Long-range distances and conformational changes

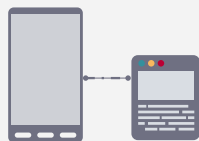




Limitations of Experimental Methods



- Different conformations
 - Same experimental data
- Cannot determine unique structural model

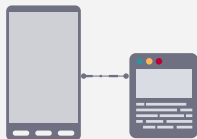




Limitations of Experimental Methods

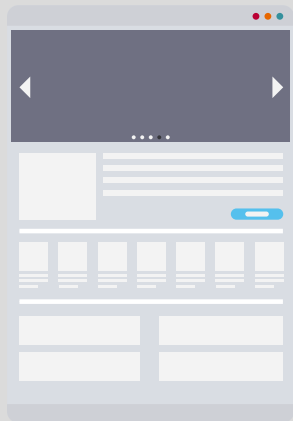
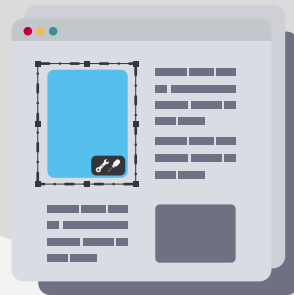
- **Ensemble averaging** → rare states missed
- **NMR** → Time-consuming and require high sample concentration
- **HDX-MS** → Low resolution
- **SAXS** → No local structural detail
- **smFRET** → Limited to labelled positions
- **Experimental data often incomplete** → poorly annotated or synthetic proteins

Thus, it motivates computational and machine learning approaches!



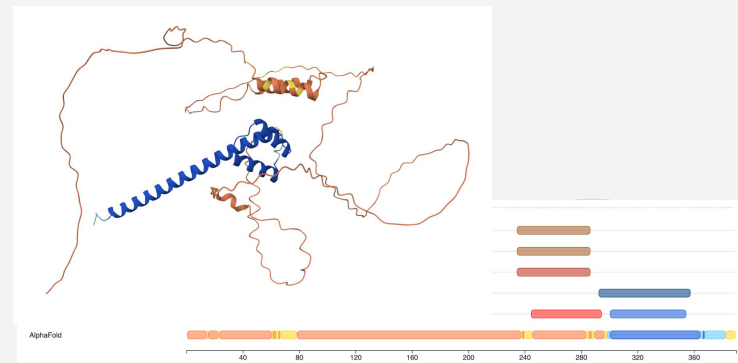
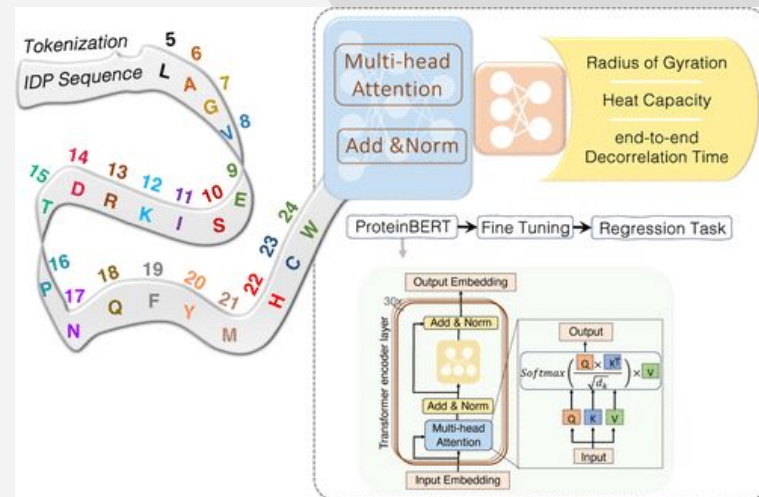
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Computational and ML approaches



Previous Computational Methods

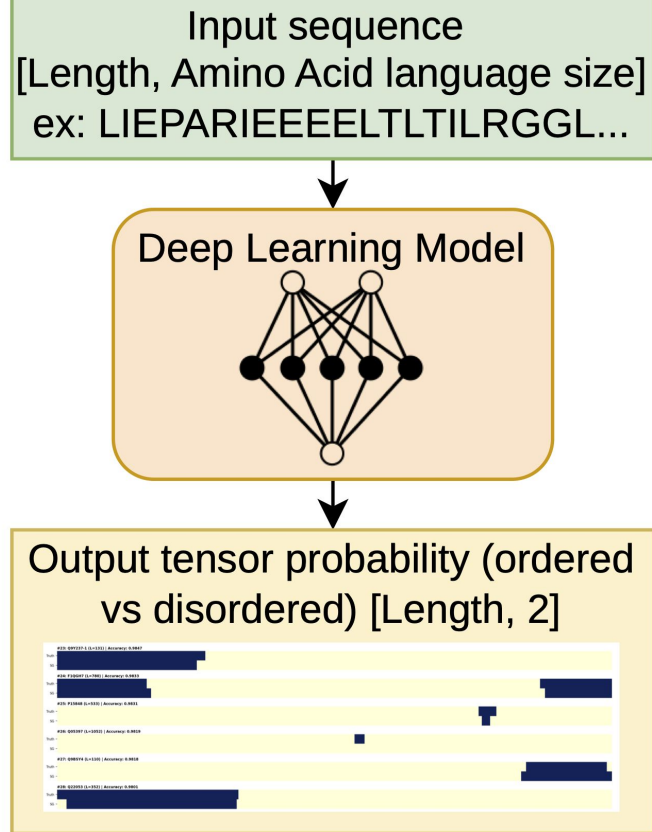
- Prediction of simulated protein features (Radius of gyration, heat capacity, decorrelation time etc.) (Mollaei et al., 2024)-> **requires MD simulation as training data**
- Other methods predict IDR's using **deep neural networks** (Hanson et al., 2019, Hou et al., 2024), but such methods require a large number of inputs (sequence alignment (PSSM-BLAST), co-evolutionary data, crystallized structure, NMR, imaging data etc.) and many of which involve **normalization (external accuracy ~0.6-0.7)**
- AlphaFold -> connotation that regions of "low-confidence" = **intrinsically disordered (high/low confidence prediction $\neq$ protein dynamics itself, just model weight confidence)**



Our method

Amino-acid resolution prediction of intrinsically disordered regions of proteins with putative disorderness characters

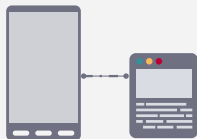
- Direct tokenized sequence-to-embedding tensor method with deep network architecture
- Each amino acid treated as individual token that is given prediction of disorderness
- Binary classification (0: ordered, 1: disordered) -> essentially predicting specific “regions” a protein is disordered



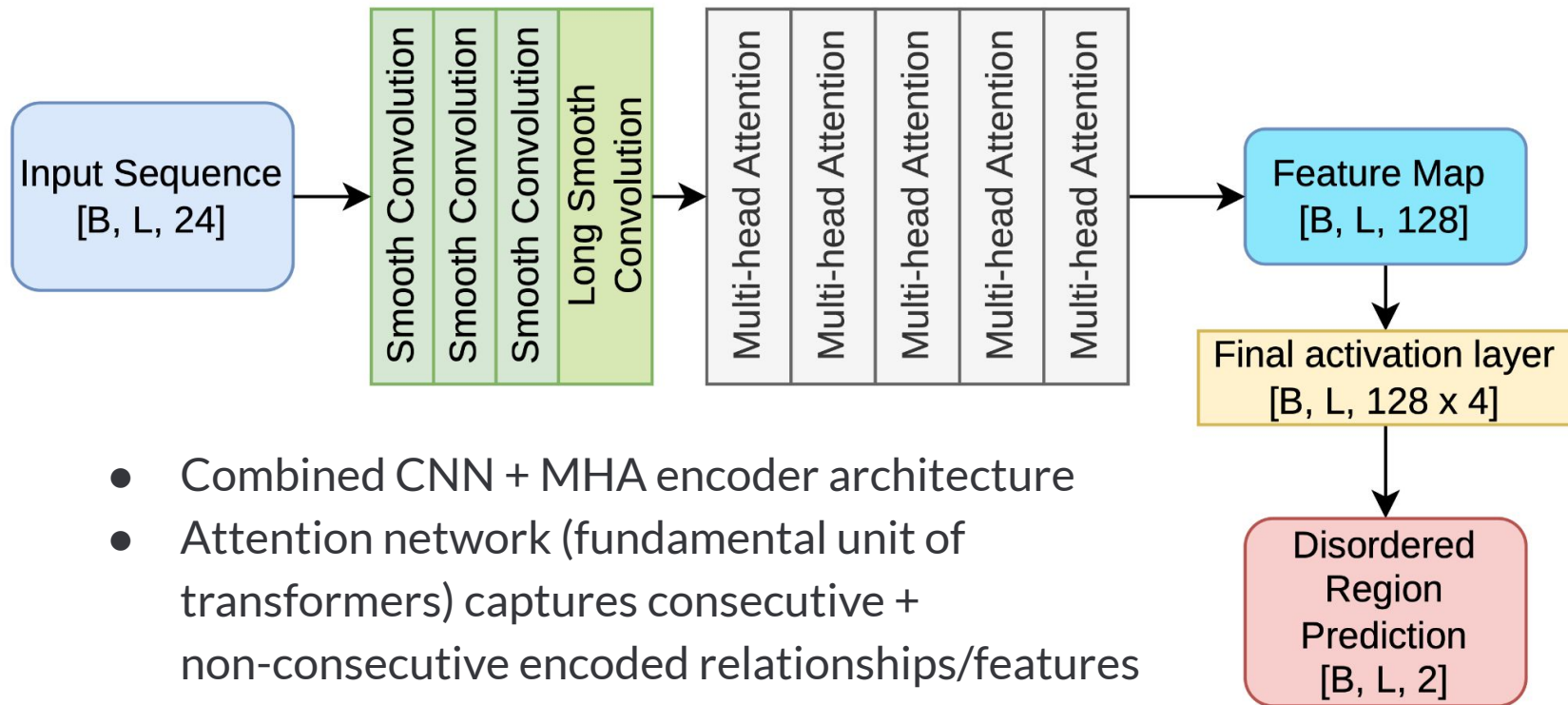
Dataset Collection



- 3 distinct databases: all open-source, data downloadable online
- **Disprot (3199 sequences)** - General databank of disordered proteins from various sources
- **LLPSDB (332 sequences)** - Liquid-liquid phase separation data (phase separation: homogenous solution -> “organelle-like” high-conc. state)
- **FuzDB (372 sequences)** - “fuzzy” region, protein-biomolecule interaction relying on disordered segments of proteins
- Data cleaning (remove duplicates, handle length mismatch): **3,901 entries**
- Other websites available (ex. *MobiDB*, *PhaSePro*, *AmyPro* etc.) but inaccessible/unuseable (maintenance, data overlap, majority of disordered annotations are computational predictions, often through AF2/AF3)
- Train/test split: 341 sequences (~1/11 of data) chosen to use as “external” validation dataset with same length distribution

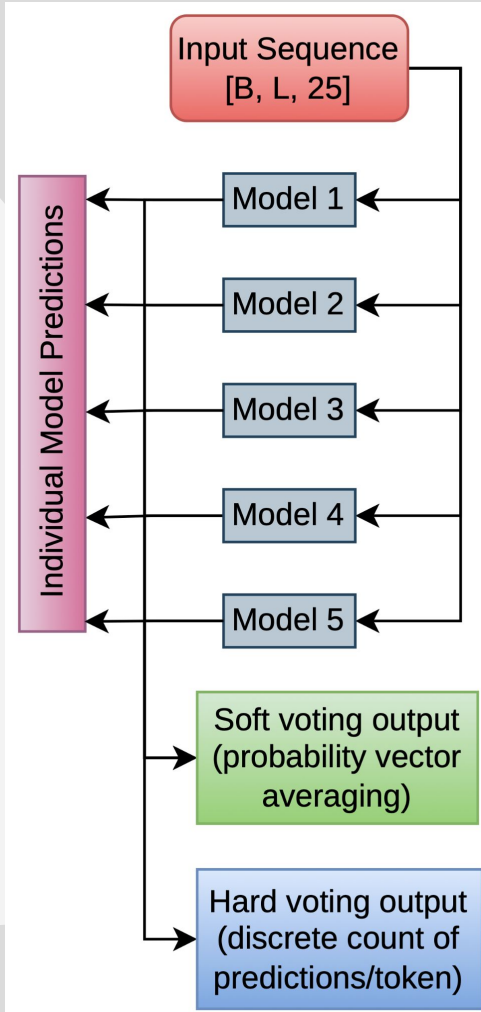
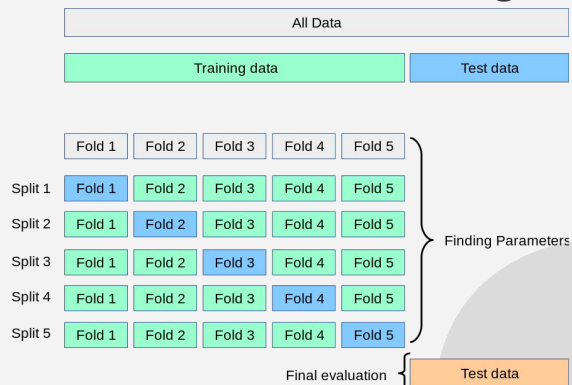


Model Architecture



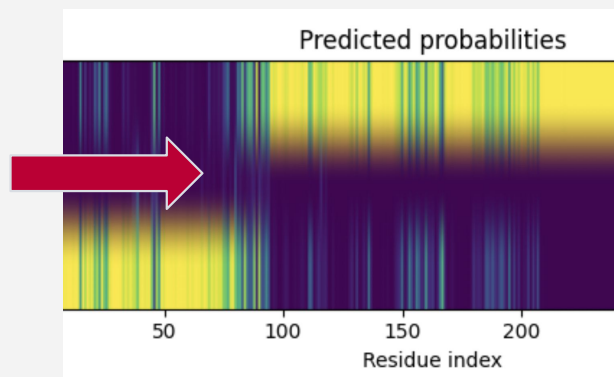
Ensemble Learning

- Prevents overfitting + boosts generalization
- Split dataset to 5 equal parts using stratified k-fold, each fold is 4 parts training + 1 part validation
- Macro-F1 scores can be used as metric (5-fold cross-validation)
- Consensus prediction made through “hard” (discrete voting of most frequent state/token) or “soft” (averaged probability vectors) voting
- Use consensus result to obtain external validation results + query predictions (voting)



Model Training

- Loss method: **Focal Loss** with **Total Variation denoising** (class imbalance + smooth transition regions between ordered and disordered) - alpha weights for focal loss determined by **binary class distribution** (0.6229, 2.5347), gamma = 2.0
- Optimizer: **Adam** (lr=2e-4, **scheduler** of 0.7 applied if validation loss does not improve)
- Number of epochs/fold: 100
- Batch size: 16



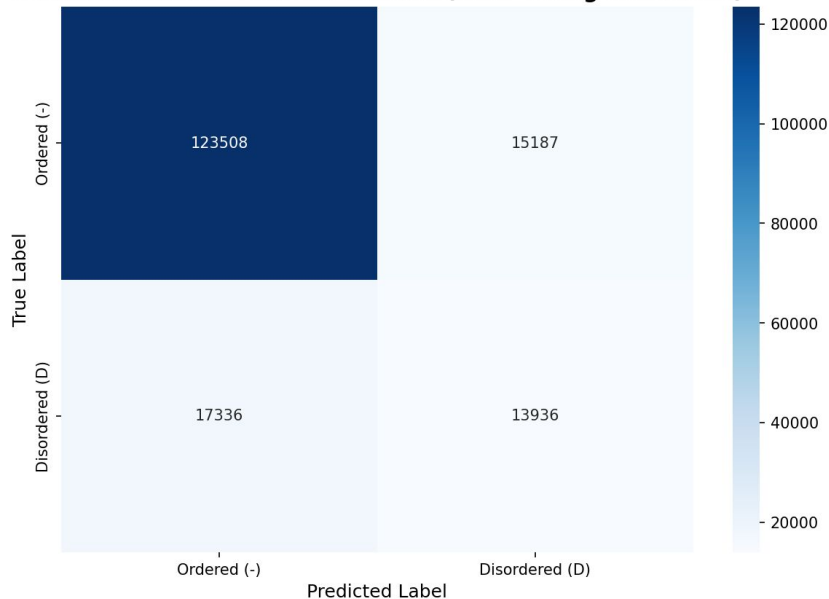
Results

- Macro-F1 score (final validation dataset): 0.67
- Hard voting outperform soft voting results

Data version: V0_1 EXTERNAL DATASET - HARD VOTING CONSENSUS (Across All 5 Folds)

	precision	recall	f1-score	support
Ordered (-)	0.88	0.89	0.88	138695
Disordered (D)	0.48	0.45	0.46	31272
accuracy			0.81	169967
macro avg	0.68	0.67	0.67	169967
weighted avg	0.80	0.81	0.81	169967

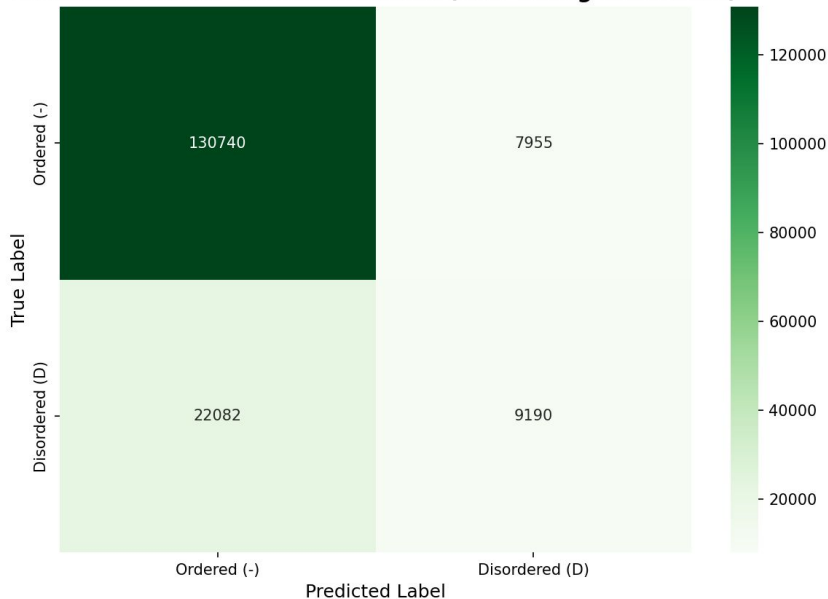
Confusion Matrix - External Dataset (Hard Voting Consensus)



EXTERNAL DATASET - SOFT VOTING CONSENSUS (Across All 5 Folds)

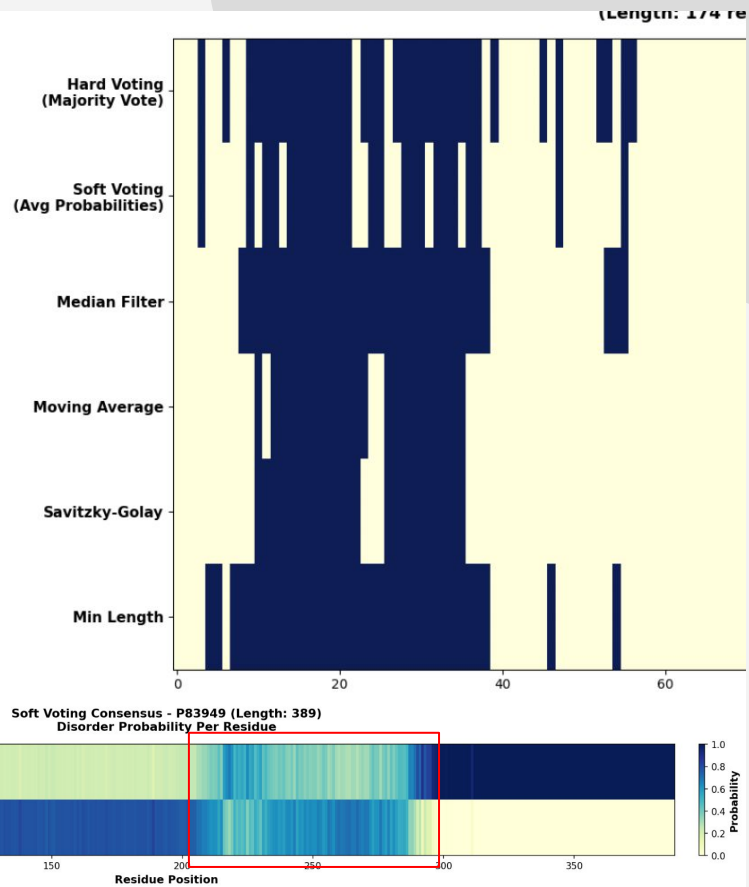
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accuracy			0.82	169967
macro avg	0.70	0.62	0.64	169967
weighted avg	0.80	0.82	0.80	169967

Confusion Matrix - External Dataset (Soft Voting Consensus)

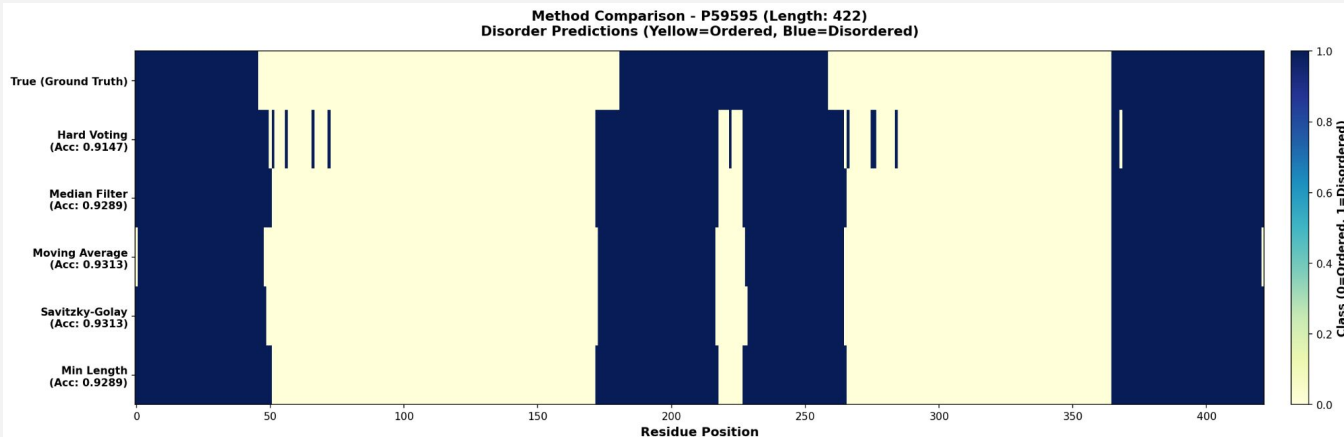
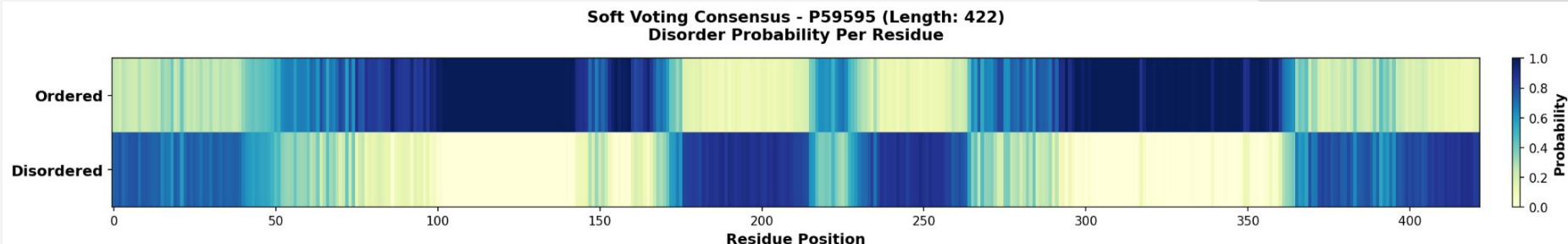


Prediction Post-Processing

- Problem with initial predictions: “intermix” issue where disorder/order tokens **alternate unnaturally at junctions**
- **Smoothing** applied to hard voting (more accurate) output for more “defined” region of where disordered region starts/stops
 - Median smoothing (hard)
 - Moving average (soft)
 - Savitzky-Golay algorithm (soft)
 - Minimum length (hard)
- Soft voting can still reveal regions model is “confident” in its prediction

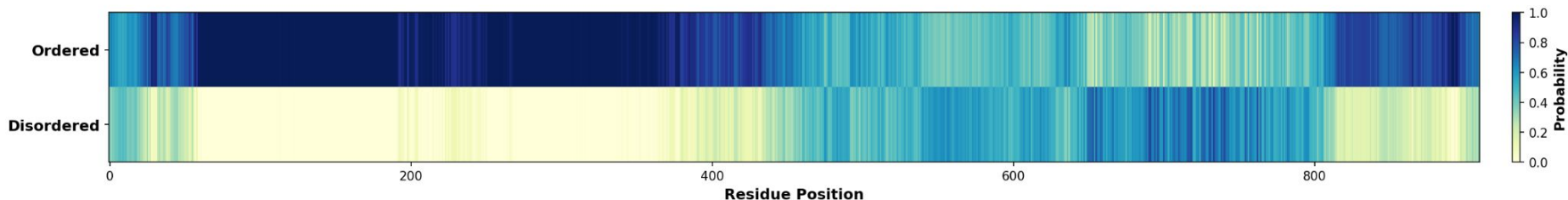


Prediction Example - P59595 (SARS Coronavirus Nucleoprotein)

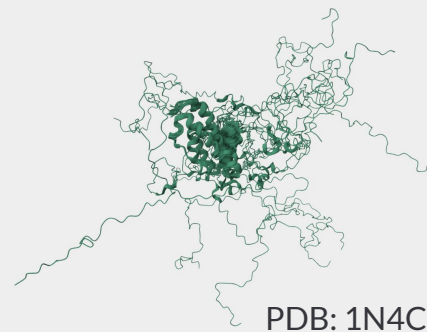
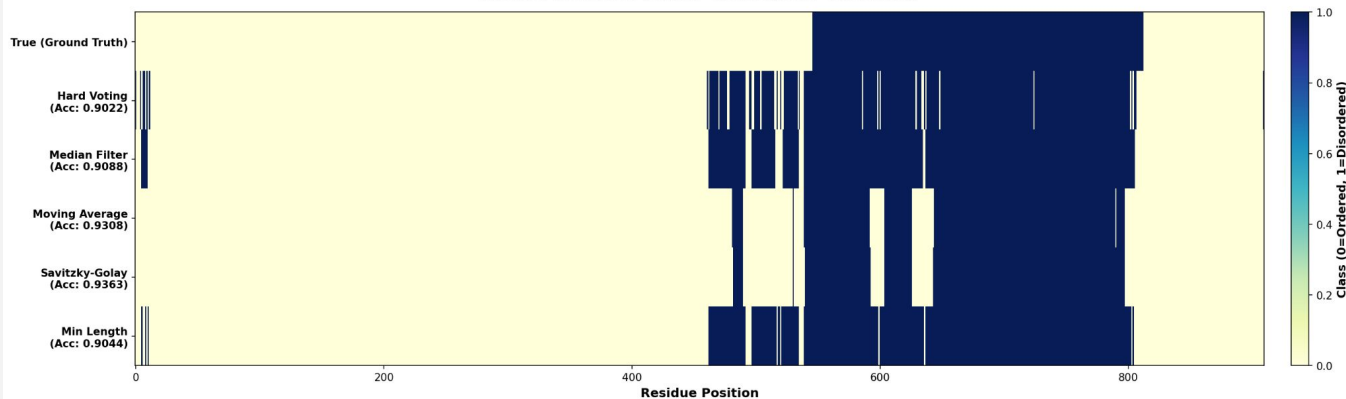


Prediction Example - Q27974 (*Bos Taurus* putative tyrosine-protein phosphatase auxilin)

Soft Voting Consensus - Q27974 (Length: 910)
Disorder Probability Per Residue



Method Comparison - Q27974 (Length: 910)
Disorder Predictions (Yellow=Ordered, Blue=Disordered)



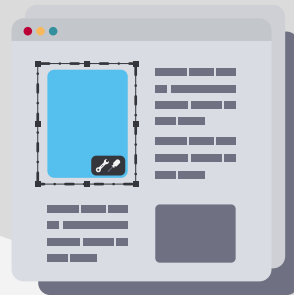
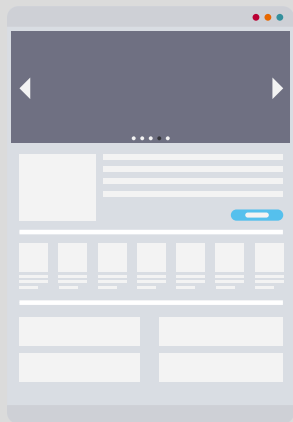
PDB: 1N4C

Advantages & Limitations

- General understanding of IDR's (~60% of validation dataset contained reasonable predictions)
- Lightweight deployment (model file 15.5 MB/fold)
- Encoded sequence-to-disorder context without reliance on evolutionary data (*low sequence complexity, high net charge, low hydrophobic content etc.*)
- Poor accuracy for several other protein classes - F1 score for disordered region is low
- Model has general context of the area that a protein sequence is disordered, but it may be off by some quantity of amino acids
- Intermixing problem still remains even with applied smoothing

04

Optimization & ESM2-powered Gradient Boosting framework



Optimization for Method 1 & 2

Loss Function (Method1)	Focal Loss: Adjusts the loss contribution of easy examples via the focusing parameter, forcing the model to prioritize the 19.8% minority class. <i>γ(Focusing Parameter): Reduces loss for easy examples.</i> <i>α(Balancing Factor): Handles the 19.8% minority class imbalance.</i> $FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$	TV Regularization: Enforces local continuity and penalizes high-frequency noise, ensuring predicted regions are biologically/physically contiguous. <i>u_i: The predicted disorder probability score for the i-th residue in the sequence.</i> $TV(u) = \sum_{i=1}^{N-1} u_{i+1} - u_i $
Metric	Macro F1: Provides an unweighted average of class-specific F1-scores, preventing bias toward the majority (ordered) class during evaluation	X Accuracy: If model guess every single amino acid is ordered to get 80.2% accuracy clearly meaningless to our research.
Sequence Smoothing	Average and Median Filtering: Suppresses isolated spikes and local variance Minimum length constraint: Ensure structural realism	Savitzky-Golay Filtering: Performs local polynomial regression to smooth the sequence

Advantage of ESM-2 & Boosting algorithm

Drawing on the state-of-the-art performance of DisPredict3.0 and DeepDRP (**ESM-2 based**) , we leverage **ESM-2** to capture the latent evolutionary patterns essential for accurate disorder prediction

Feature	Bagging (Random Forest)	Boosting (XGB/ LGBM/Cat)
Architecture	<u>Parallel</u> Independent models	<u>Sequential</u> Iterative learning
Learning Goal	<u>Stability</u> Reduces Variance	<u>Accuracy</u> Reduces Bias
Weighting	<u>Uniform</u> All trees are equal	<u>Weighted</u> Best trees carry more power

Boosting Algorithm	Core Innovation	Key Advantage
XGBoost	Sparsity-Aware Learning	Efficiency: A scalable workhorse that handles high-dimensional embedding vectors with extreme resource efficiency.
LightGBM	GOSS (Gradient-based One-Side Sampling)	Speed: Accelerates processing of massive datasets by focusing on samples with larger gradients without losing precision.
CatBoost	Ordered Boosting	Robustness: Eliminates "prediction shift" and target leakage, ensuring the highest empirical quality and generalization.

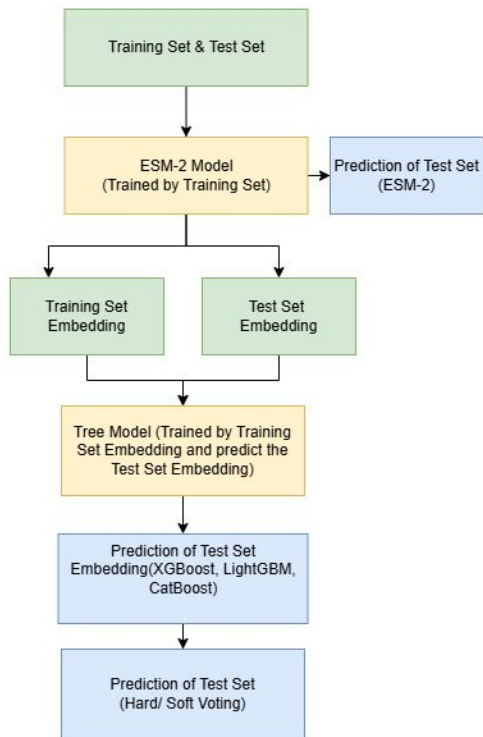
Method2: Hybrid Approach Workflow & Result

Why This Hybrid Approach Works

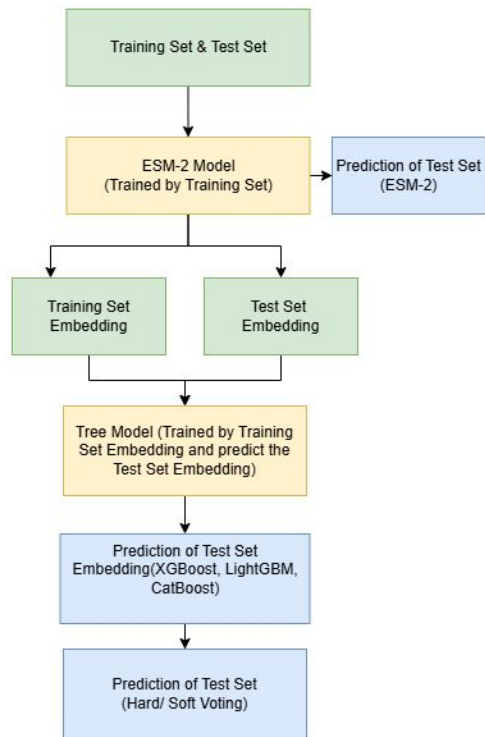
- **Biological Synergy:** Combines ESM-2's deep evolutionary "grammar" with Tree models' superior ability to handle non-linear tabular data.
- **Enhanced Precision:** Gradient Boosting Trees (XGBoost/CatBoost) refine raw LLM outputs, creating sharper decision boundaries for Order vs. Disorder.
- **Automated Feature Engineering:** Eliminates manual calculation of physical properties; embeddings capture complex latent biological features automatically.

Limitation

- **Supervised Embedding Bias:** Since ESM-2 was fine-tuned on the training set, internal 5-fold CV scores may be overly optimistic due to label-aware feature extraction.
- **Independent Validation:** To ensure zero data leakage, final performance claims are strictly based on a Blind External Test Set, providing a true measure of generalization on unseen sequences.



Method2: Hybrid Approach Workflow & Result



Model	5-fold CV Macro F1 score	Test Macro F1 score	Threshold
ESM2		0.67	-
XGBoost	0.731	0.70	0.66
LightGBM	0.737	0.70	0.68
CatBoost	0.759	0.70	0.67
Soft Voting	-	0.70	0.68
Hard Voting	-	0.70	-

Optimization

- **Threshold Searching:** Identify the classification threshold for test set that maximizes the Macro F1-score across 5-fold cross-validation .
- **Bayesian Optimization (Hyperopt):** Efficiently explores high-dimensional continuous search spaces to optimize tree model hyperparameter.

Results

Method 1: Neural Network

=====

Data version: V0_1 EXTERNAL DATASET - HARD VOTING CONSENSUS (Across All 5 Folds)

=====

	precision	recall	f1-score	support
Ordered (-)	0.88	0.89	0.88	138695
Disordered (D)	0.48	0.45	0.46	31272
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weighted avg	0.80	0.81	0.81	169967

=====

EXTERNAL DATASET - SOFT VOTING CONSENSUS (Across All 5 Folds)

=====

	precision	recall	f1-score	support
Ordered (-)	0.86	0.94	0.90	138695
Disordered (D)	0.54	0.29	0.38	31272
accuracy			0.82	169967
macro avg	0.70	0.62	0.64	169967
weighted avg	0.80	0.82	0.80	169967

While Method 1 serves as a powerful original architecture capable of learning order-disorder transitions directly from raw data without external embeddings, Method 2 Hybrid Framework demonstrates how leveraging ESM-2 can further push the boundaries of performance.

By integrating these deep evolutionary patterns with tree-based ensembles, we achieved a **4.5%** improvement in Macro F1-score, peaking at **0.70**

Method 2: Hybrid Approach

 XGBoost Optimized Threshold: 0.66

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XGBOOST - FINAL EXTERNAL PERFORMANCE

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	precision	recall	f1-score	support
Ordered	0.89	0.89	0.89	138695
Disordered	0.51	0.51	0.51	31270
accuracy			0.82	169965
macro avg	0.70	0.70	0.70	169965
weighted avg	0.82	0.82	0.82	169965

 Optimal CatBoost Threshold: 0.67

 Highest Train Macro F1: 0.7569

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CATBOOST - FINAL EXTERNAL PERFORMANCE (Threshold: 0.67)

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	precision	recall	f1-score	support
Ordered	0.89	0.87	0.88	138695
Disordered	0.48	0.55	0.51	31270
accuracy			0.81	169965
macro avg	0.69	0.71	0.70	169965
weighted avg	0.82	0.81	0.81	169965

 LGBM Optimized Threshold: 0.68

	precision	recall	f1-score	support
0	0.89	0.88	0.89	138695
1	0.50	0.53	0.51	31270
accuracy			0.81	169965
macro avg	0.70	0.71	0.70	169965
weighted avg	0.82	0.81	0.82	169965

Method 2: Hybrid Approach (Ensemble Learning of Tree Model)

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ENSEMBLE: HARD VOTING

=====

	precision	recall	f1-score	support
Ordered	0.89	0.88	0.89	138695
Disordered	0.50	0.53	0.52	31270
accuracy			0.82	169965
macro avg	0.70	0.71	0.70	169965
weighted avg	0.82	0.82	0.82	169965

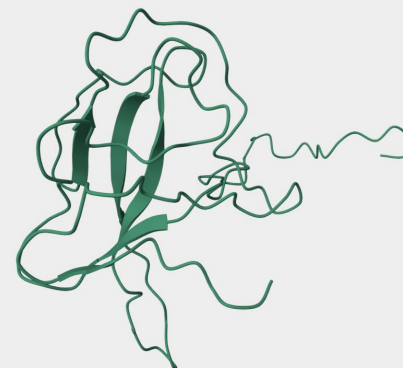
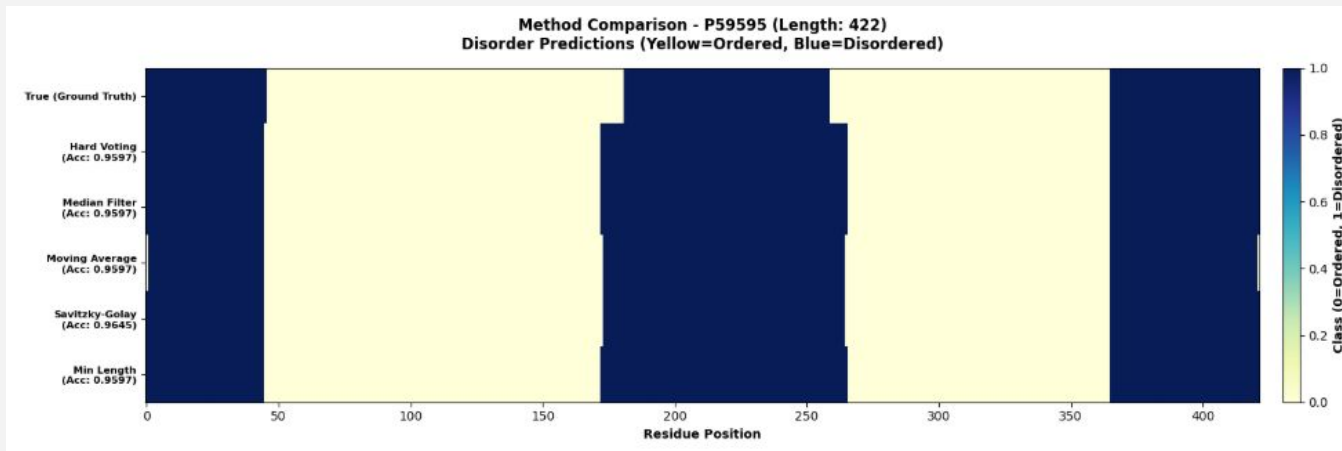
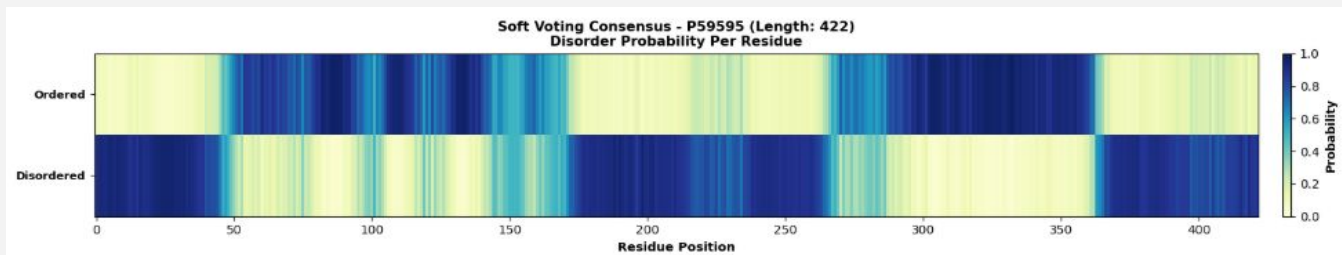
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ENSEMBLE: SOFT VOTING (Threshold: 0.68)

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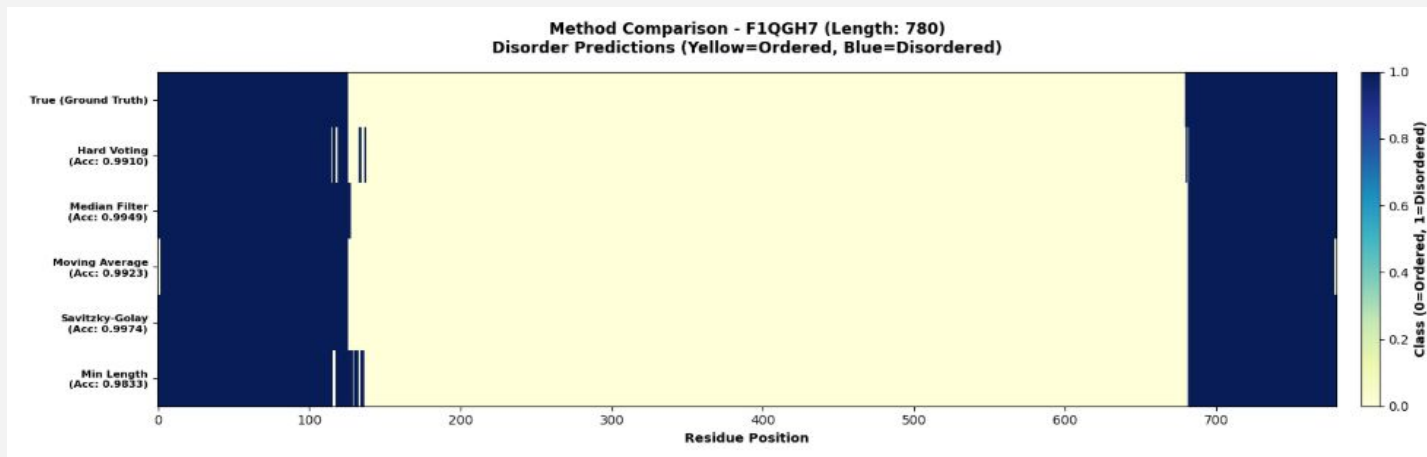
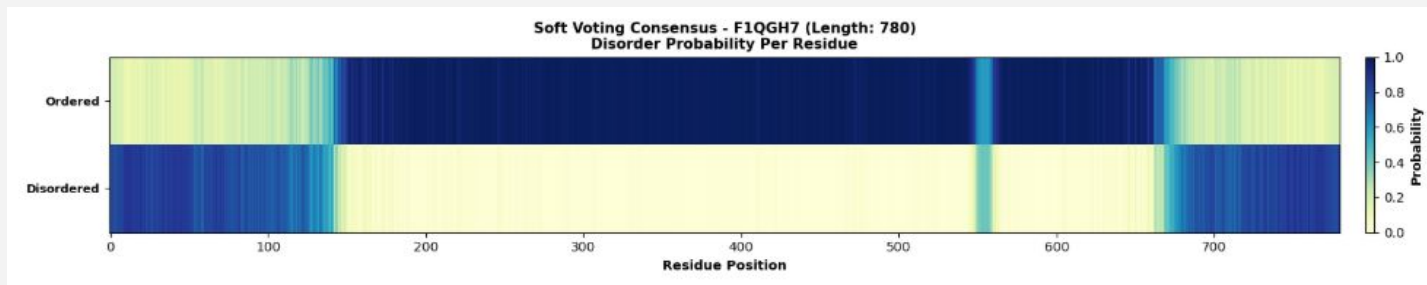
	precision	recall	f1-score	support
Ordered	0.89	0.89	0.89	138695
Disordered	0.52	0.51	0.52	31270
accuracy			0.82	169965
macro avg	0.71	0.70	0.70	169965
weighted avg	0.82	0.82	0.82	169965

Prediction Example - P59595 (SARS Coronavirus Nucleoprotein)



PDB: 1SSK

Prediction Example - F1QGH7 Catenin beta-1



PDB: 2Z6G

Advantage & Limitation

Key Advantages

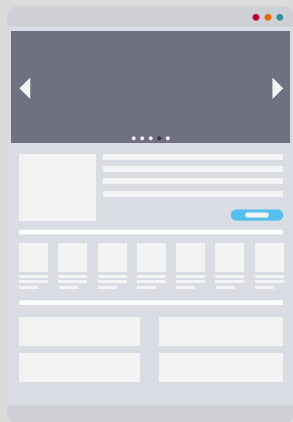
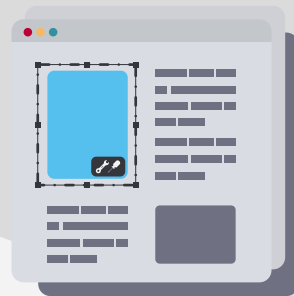
- **Robust Imbalance Handling:** Tree models effectively manage the skewed ratio of disordered vs. ordered residues common in biological datasets.
- **Modular "Pluggable" Architecture:** High flexibility to swap the encoder (ESM-2, ProtT5, Ankh) or the classifier (RF, SVM, MLP) based on specific structural targets.
- **Automated Feature Extraction:** Eliminates manual feature engineering by leveraging ESM-2's deep evolutionary grammar.

Limitation

- **Supervised Embedding Bias:** Since ESM-2 was fine-tuned on the training set, internal 5-fold CV scores may be overly optimistic due to label-aware feature extraction.
- **Independent Validation:** To ensure zero data leakage, final performance claims are strictly based on a Blind External Test Set, providing a true measure of generalization on unseen sequences.

05

Challenges and Future Outlook



Challenges: Data Quality

- “Ground Truth” Ambiguity

The definition challenge:

The term "disordered" lacks a standardized definition across different datasets and experimental frameworks.

What was once a binary label is now recognized as a complex spectrum of states, leading to vague and loosely used terminology (Dunker et al., 2013).

The "Consistency Gap"

Experimental results for the same protein often disagree due to these differing definitions (Shi, Peng et al., 2019).

This gap creates significant hurdles in building reliable IDP databases and computational models.

Challenges: Data Quality

- “Ground Truth” Ambiguity

X-Ray Crystallography	NMR Analysis
• Labels residues that do not show up in a crystal structure as "disordered" (Zanotti, 2023). • Focuses on the lack of a fixed coordinate in the crystal lattice.	• Identifies overlapping signals between "highly flexible" and "completely disordered" portions (Felli & Pierattelli, 2015). • Provides a more dynamic view but leads to labelling discrepancies.

The same protein can be labelled differently depending on whether structural invisibility or dynamic flexibility is measured.

Challenges: Nature of IDPs

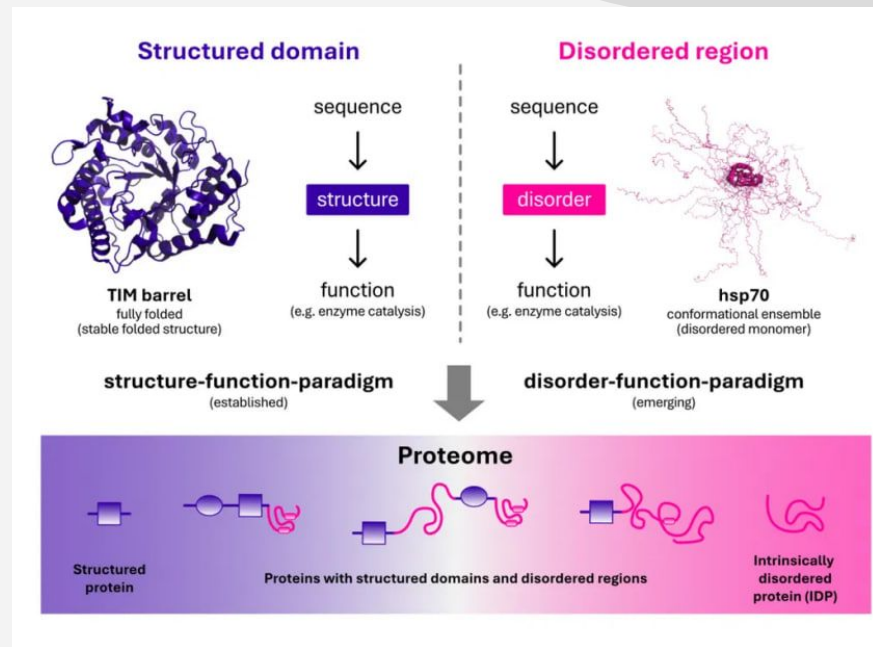
- Environmental and Contextual Sensitivity

The "Shapeshifter" Problem

Conformational Ensembles -

IDPs do not exist in a single fixed state. Instead, they exist as conformational ensembles, constantly shifting their structure (Chakravarty et al., 2023).

The Limitation: Current models often hinge on sequence data alone, overlooking how IDPs behave within the actual complexity of a living cell.

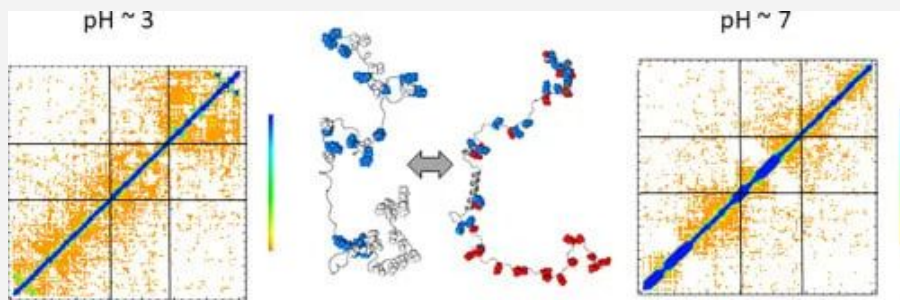


Challenges: Nature of IDPs

- Environmental and Contextual Sensitivity

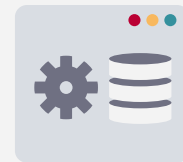
Environmental Factors: pH & Crowding

pH Shifts & Stability	Molecular Crowding
IDPs are charged molecules stabilized by specific residues. pH shifts neutralize these charges, triggering a sudden hydrophobic collapse (Lindsay et al., 2021).	High concentrations of cellular molecules create "exclusion volumes", physically pushing IDPs into more compact, folded shapes (Cino et al., 2012).



Proteins disordered at physiological pH 7.4 may be ordered in actual cellular conditions, and vice versa.

Future outlooks: Cancer research - Targeting the “Undruggable”

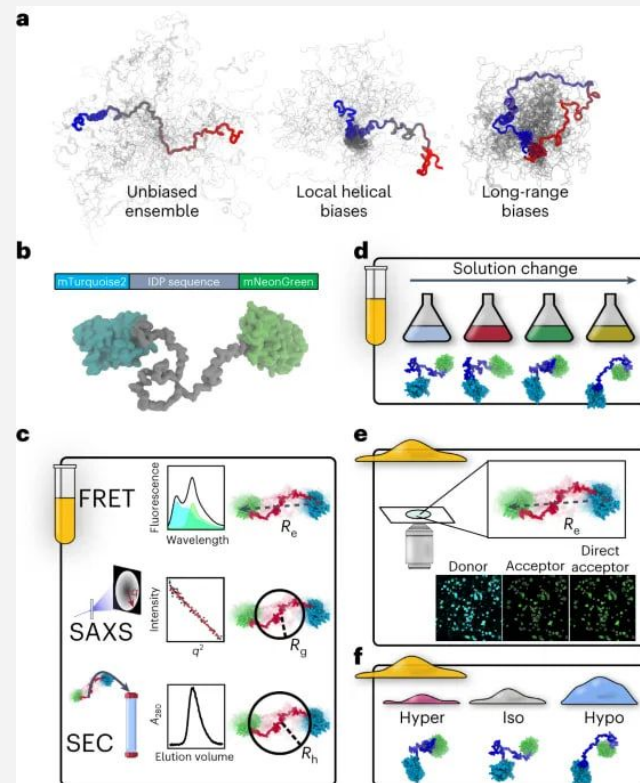


Targeting Molecular Recognition Features (MoRFs)

MoRFs are short disordered segments that undergo a disorder-to-order transition upon binding to a partner protein.

Key cancer drivers like p53 and c-Myc utilize these features to interact with multiple signaling partners.

By identifying these specific interaction motifs, our model can pinpoint the exact segments that trigger cancer-promoting activities.



Future outlooks: Cancer research

- Targeting the “Undruggable”



Cancer proteins like c-Myc, p53, and EWS-FLI1 lack a stable "lock" structure, making it difficult for traditional drug design.

By identifying MoRFs – short segments that stabilize upon binding—companies can design specific inhibitors.

Small Molecule Inhibitors or Peptidomimetics can block these specific interaction motifs, effectively "switching off" the cancer driver.

Future outlooks: Neurodegenerative diseases

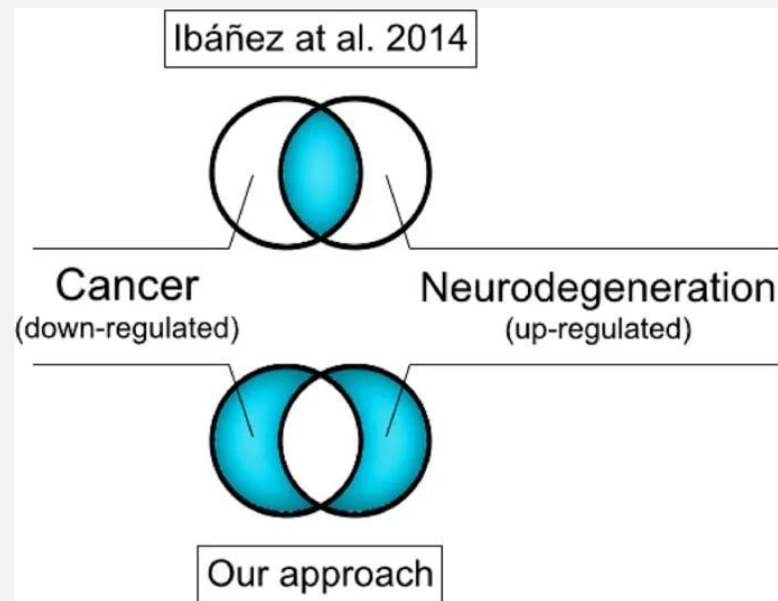
- Preventing the “clump”



Protein Aggregation

Most neurodegenerative diseases result from protein aggregation, where healthy flexible proteins turn into toxic, solid-like clumps (Pedersen et al., 2013).

These proteins naturally exist in a disordered state but clump together over time, causing irreversible cellular damage.



Future outlooks: Neurodegenerative diseases – Preventing the “clump”

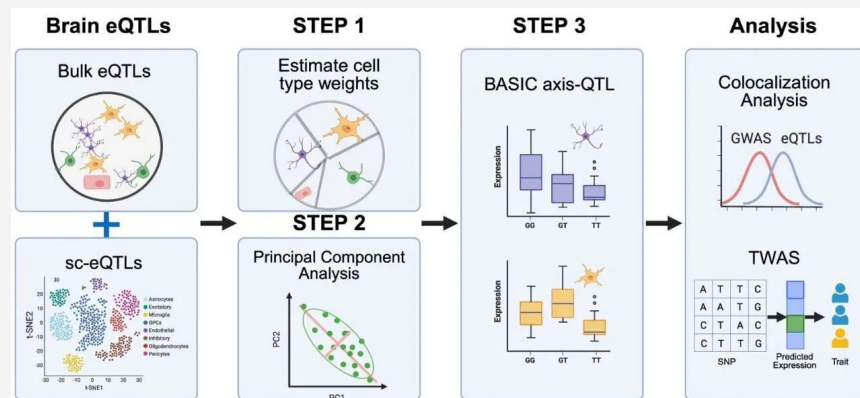


Pinpointing Triggers

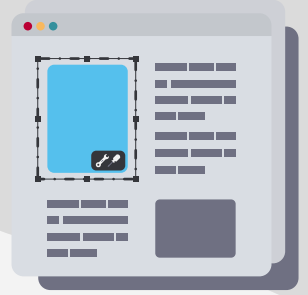
Advanced models allow for pinpointing the exact amino acids within disordered proteins that trigger the transition into a toxic clump.

Early Stabilization

Early prediction enables the stabilization of these specific residues, preventing or slowing down the clumping process (KeAi Communications, 2024).



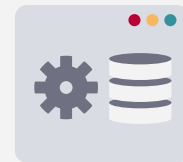
The Goal: Moving from reactive treatment to proactive prevention by identifying and neutralizing aggregation triggers at the molecular level.



Thank You!

Any Questions?

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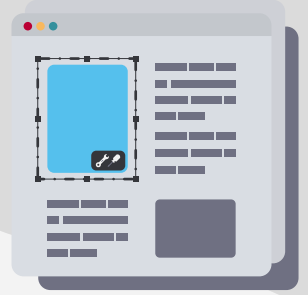
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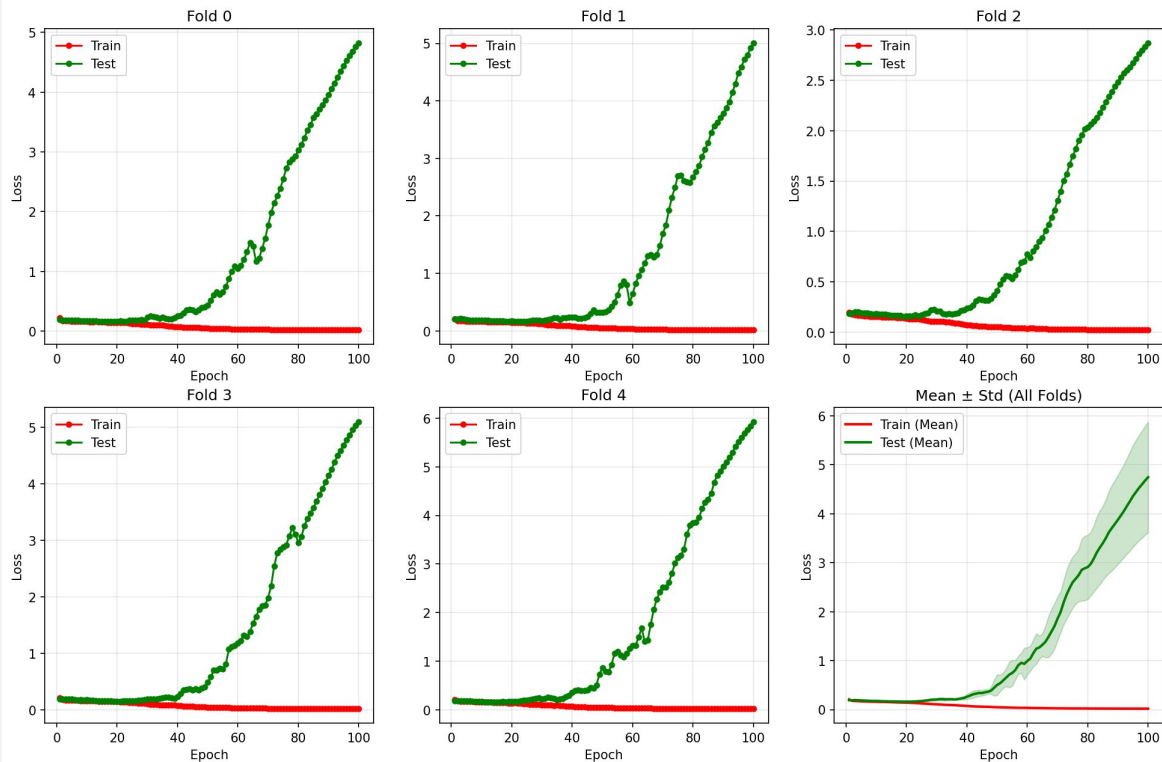
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Appendix



Consensus Loss Curves

5-Fold Cross-Validation Training Results (Model V0_1)

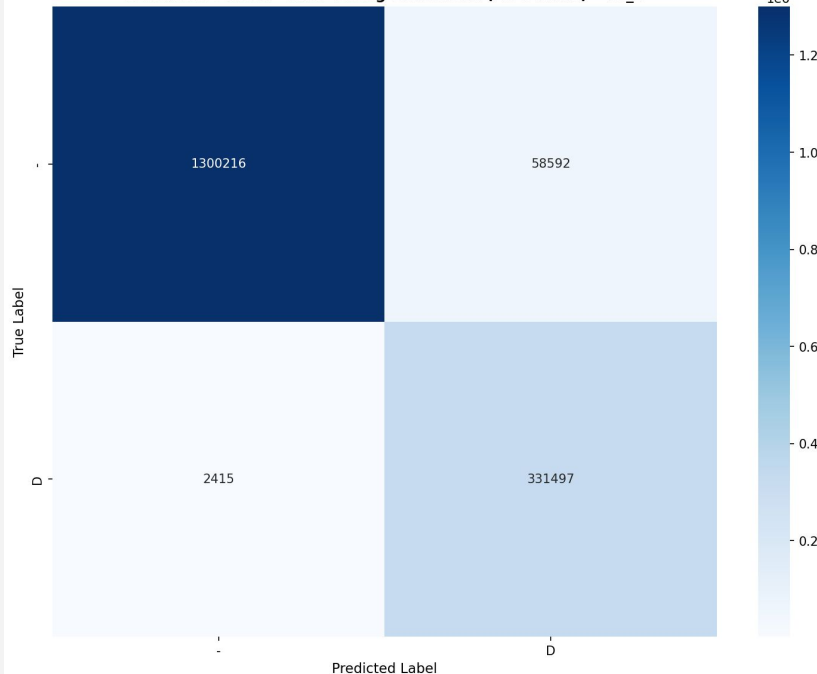


Results (Training Dataset Consensus Matrices)

HARD VOTING CONSENSUS RESULTS (Across All 5 Folds)

	precision	recall	f1-score	support
-	1.00	0.96	0.98	1358808
D	0.85	0.99	0.92	333912
accuracy			0.96	1692720
macro avg	0.92	0.97	0.95	1692720
weighted avg	0.97	0.96	0.96	1692720

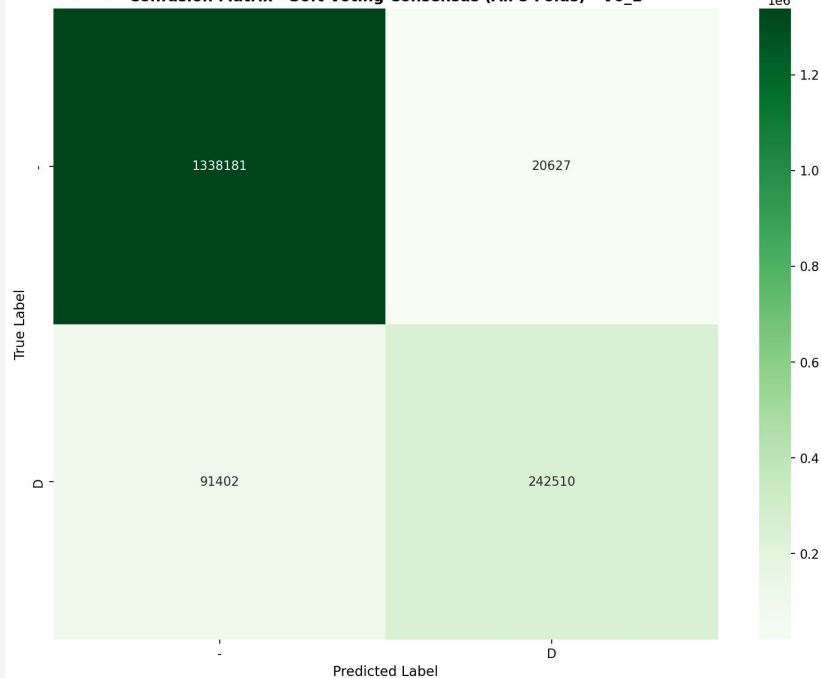
Confusion Matrix - Hard Voting Consensus (All 5 Folds) - V0_1



SOFT VOTING CONSENSUS RESULTS (Across All 5 Folds)

	precision	recall	f1-score	support
-	0.94	0.98	0.96	1358808
D	0.92	0.73	0.81	333912
accuracy			0.93	1692720
macro avg	0.93	0.86	0.89	1692720
weighted avg	0.93	0.93	0.93	1692720

Confusion Matrix - Soft Voting Consensus (All 5 Folds) - V0_1



Results

Method 1: Neural Network

=====

Data version: V0_1 EXTERNAL DATASET - HARD VOTING CONSENSUS (Across All 5 Folds)

=====

	precision	recall	f1-score	support
Ordered (-)	0.88	0.89	0.88	138695
Disordered (D)	0.48	0.45	0.46	31272
accuracy			0.81	169967
macro avg	0.68	0.67	0.67	169967
weighted avg	0.80	0.81	0.81	169967

=====

EXTERNAL DATASET - SOFT VOTING CONSENSUS (Across All 5 Folds)

=====

	precision	recall	f1-score	support
Ordered (-)	0.86	0.94	0.90	138695
Disordered (D)	0.54	0.29	0.38	31272
accuracy			0.82	169967
macro avg	0.70	0.62	0.64	169967
weighted avg	0.80	0.82	0.80	169967

Method 2: Hybrid Approach

 XGBoost Optimized Threshold: 0.66

=====

XGBOOST - FINAL EXTERNAL PERFORMANCE

=====

	precision	recall	f1-score	support
Ordered	0.89	0.89	0.89	138695
Disordered	0.51	0.51	0.51	31270
accuracy			0.82	169965
macro avg	0.70	0.70	0.70	169965
weighted avg	0.82	0.82	0.82	169965

 Optimal CatBoost Threshold: 0.67

 Highest Train Macro F1: 0.7569

=====

CATBOOST - FINAL EXTERNAL PERFORMANCE (Threshold: 0.67)

=====

	precision	recall	f1-score	support
Ordered	0.89	0.87	0.88	138695
Disordered	0.48	0.55	0.51	31270
accuracy			0.81	169965
macro avg	0.69	0.71	0.70	169965
weighted avg	0.82	0.81	0.81	169965

 LGBM Optimized Threshold: 0.68

	precision	recall	f1-score	support
0	0.89	0.88	0.89	138695
1	0.50	0.53	0.51	31270
accuracy			0.81	169965
macro avg	0.70	0.71	0.70	169965
weighted avg	0.82	0.81	0.82	169965

Method 2: Hybrid Approach (Ensemble Learning of Tree Model)

=====

ENSEMBLE: HARD VOTING

=====

	precision	recall	f1-score	support
Ordered	0.89	0.88	0.89	138695
Disordered	0.50	0.53	0.52	31270
accuracy			0.82	169965
macro avg	0.70	0.71	0.70	169965
weighted avg	0.82	0.82	0.82	169965

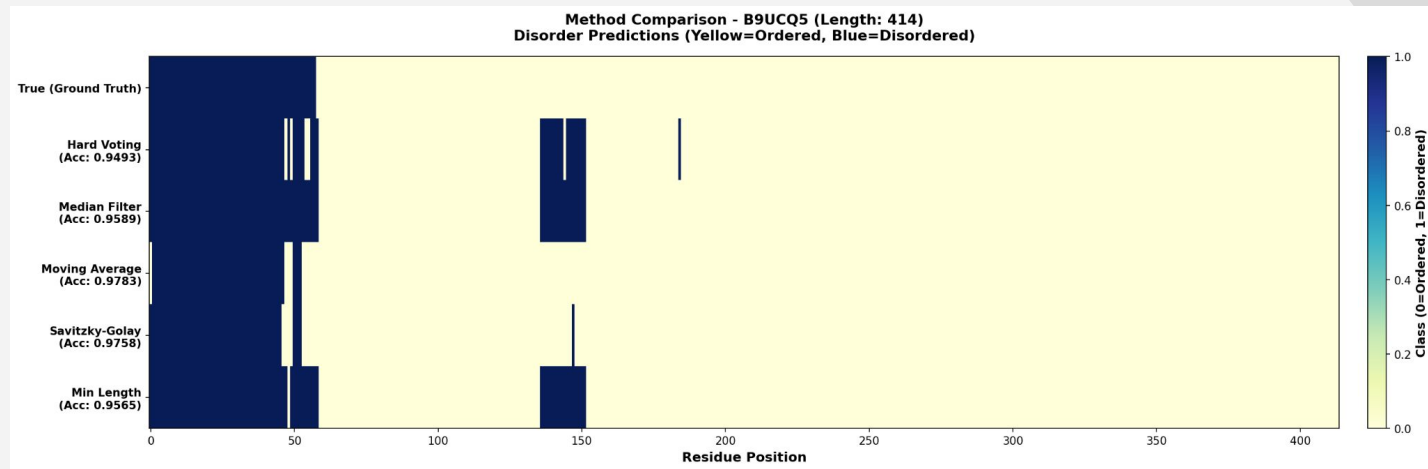
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ENSEMBLE: SOFT VOTING (Threshold: 0.68)

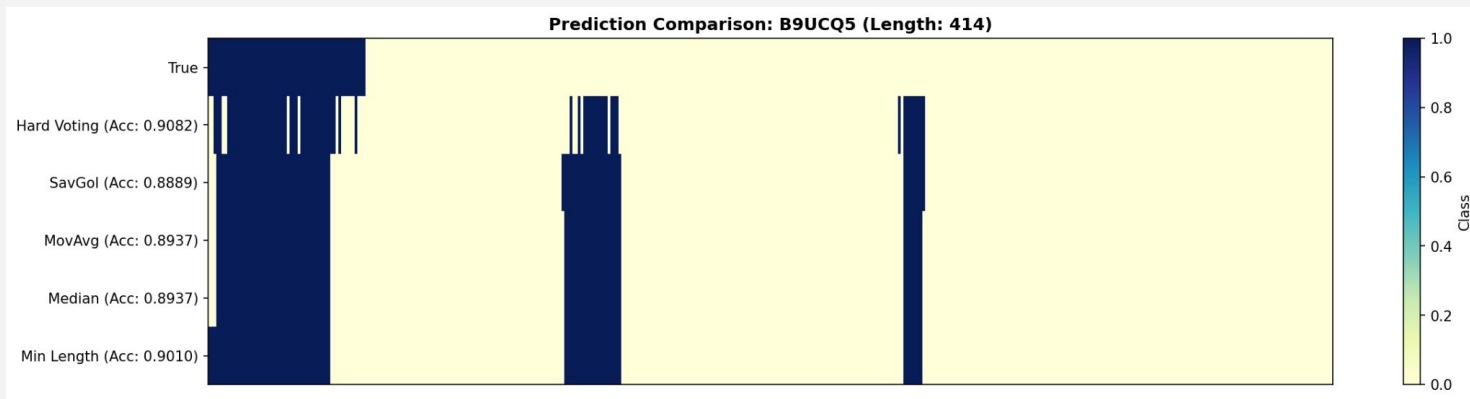
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	precision	recall	f1-score	support
Ordered	0.89	0.89	0.89	138695
Disordered	0.52	0.51	0.52	31270
accuracy			0.82	169965
macro avg	0.71	0.70	0.70	169965
weighted avg	0.82	0.82	0.82	169965

Comparison: B9UCQ5 (top: direct method, bottom: ESM2 embedding method)

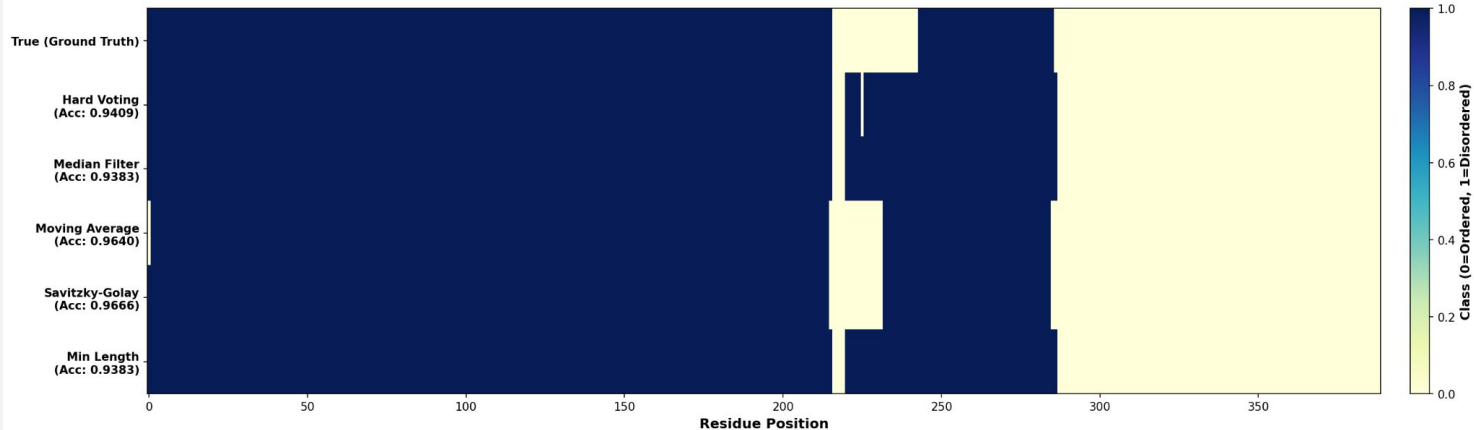


Protein ultraspiracle
homolog (Cotton
bollworm)



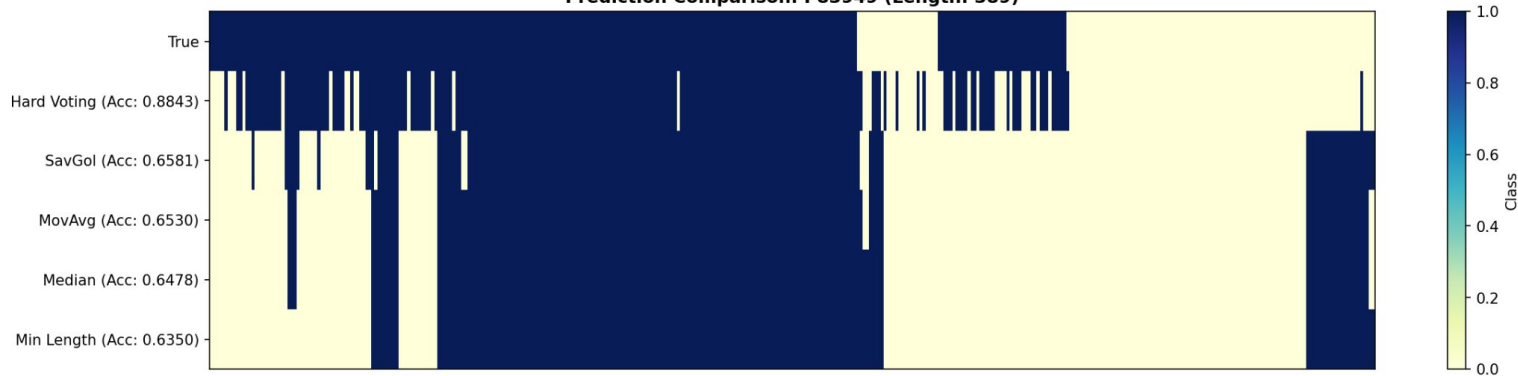
Comparison: P83949

Method Comparison - P83949 (Length: 389)
Disorder Predictions (Yellow=Ordered, Blue=Disordered)



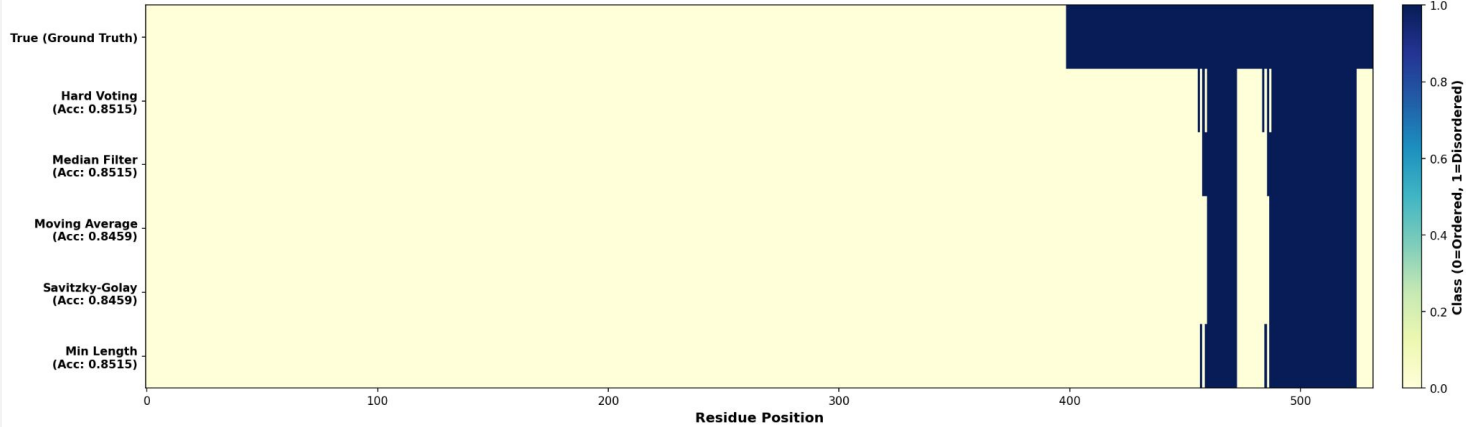
Homeotic protein
ultrabithorax (Fruit Fly)
- DNA binding

Prediction Comparison: P83949 (Length: 389)



Comparison: Q9IK92

Method Comparison - Q9IK92 (Length: 532)
Disorder Predictions (Yellow=Ordered, Blue=Disordered)

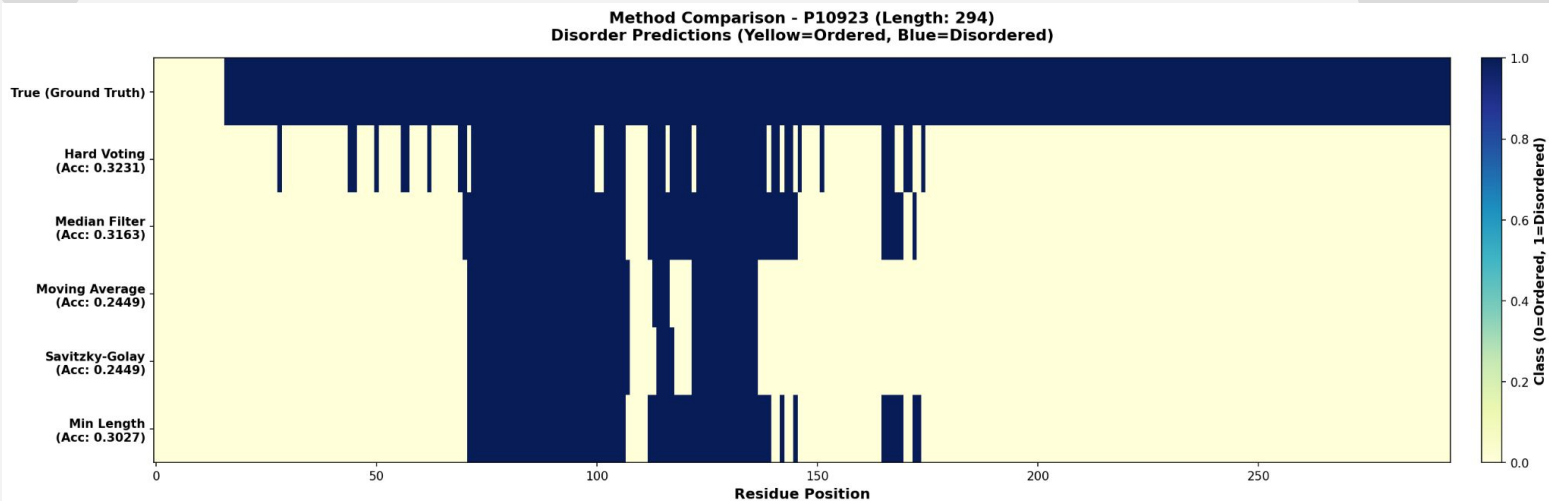


Nucleoprotein (Nipah Virus)

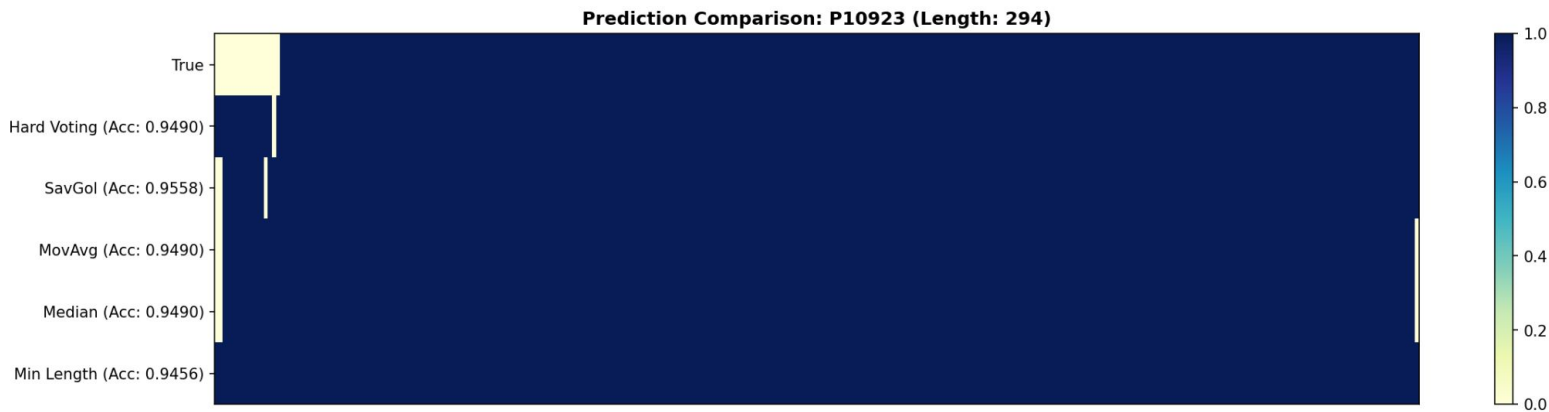
Prediction Comparison: Q9IK92 (Length: 532)



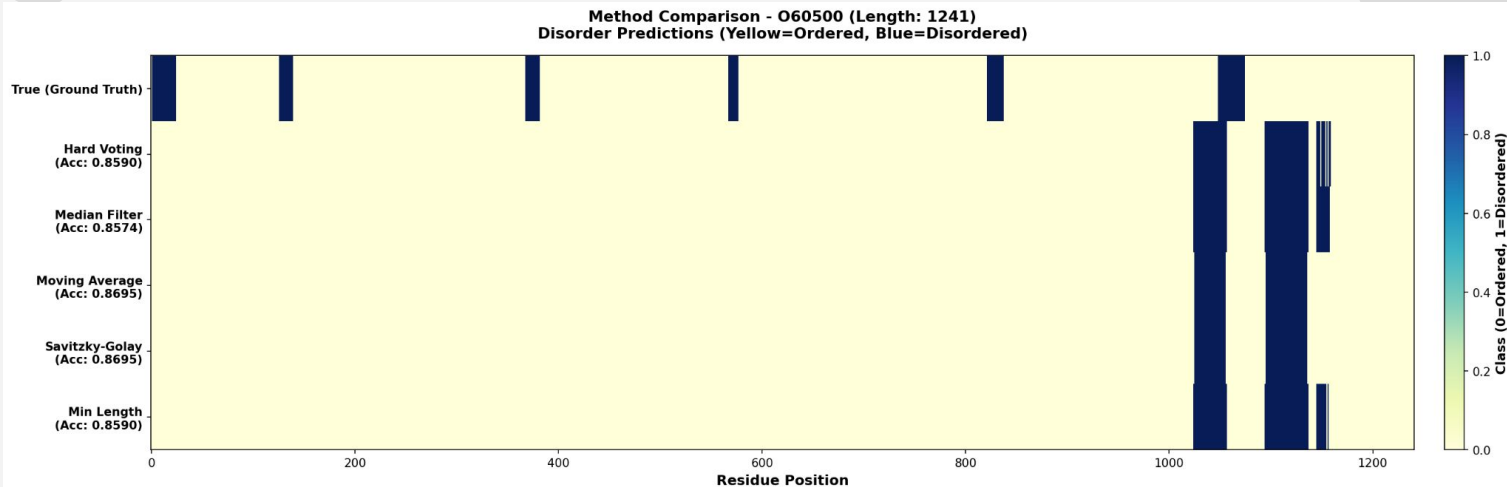
Comparison: P10923



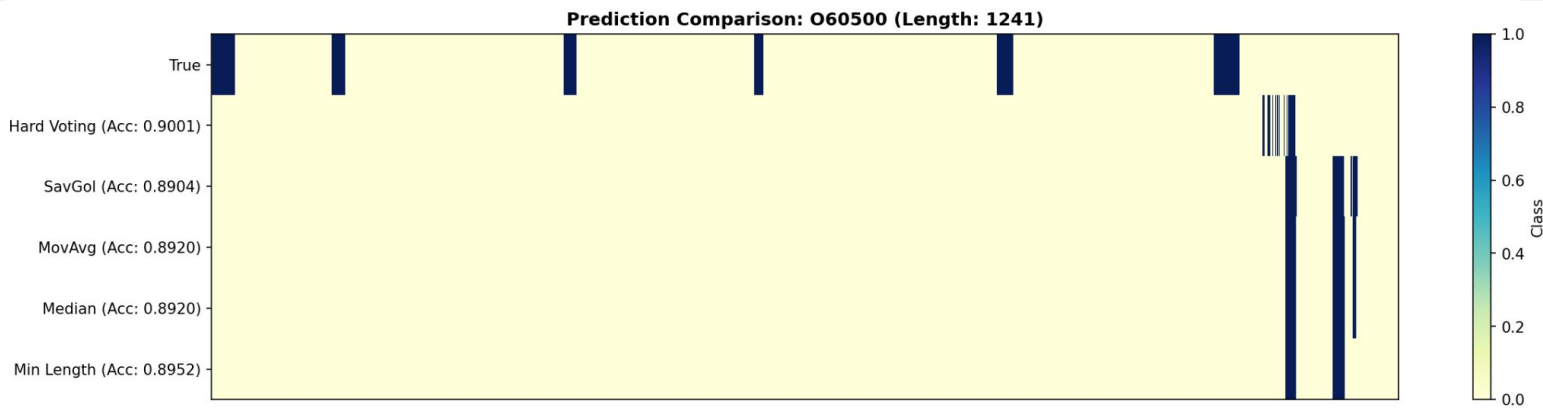
Osteopontin (mouse)



Comparison: O60500

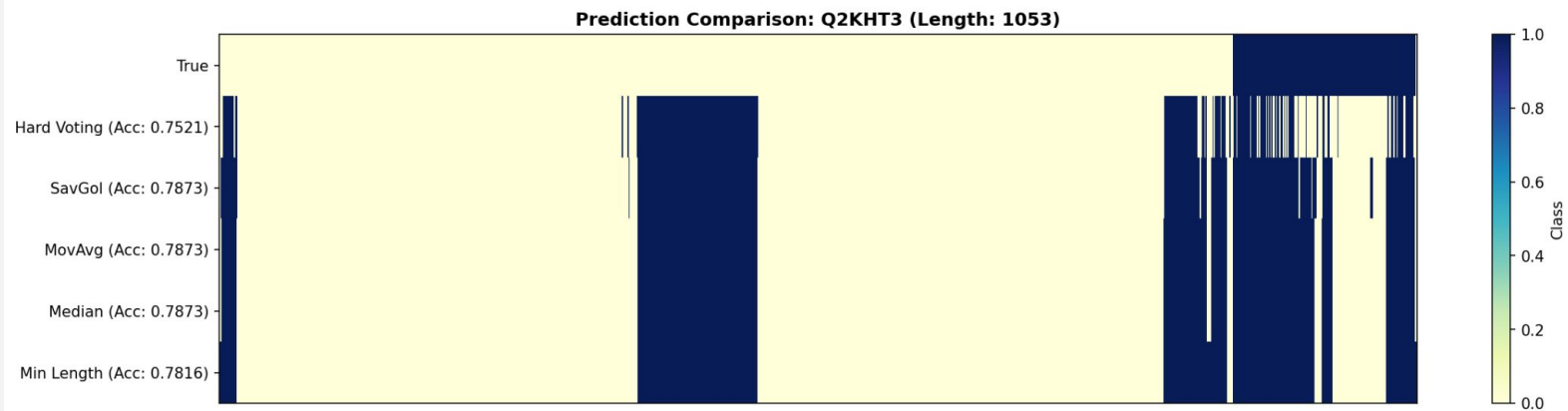
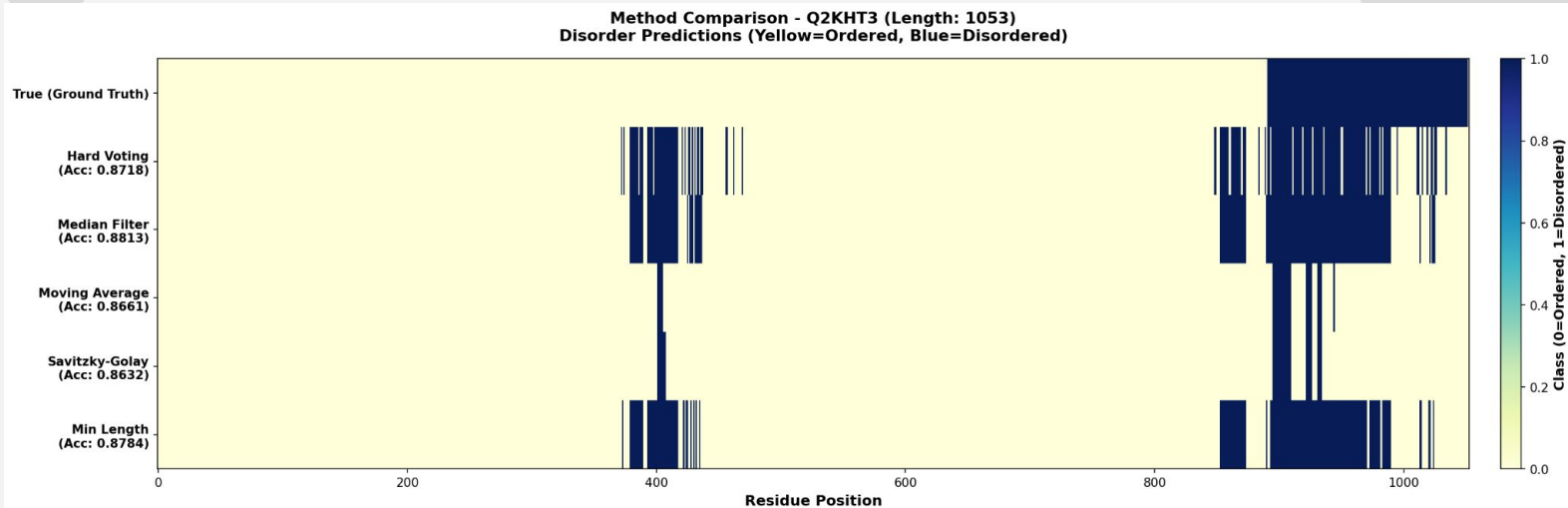


Nephrin (Human)

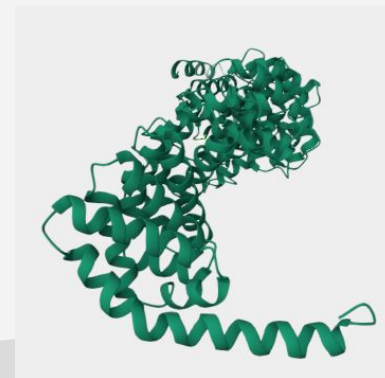
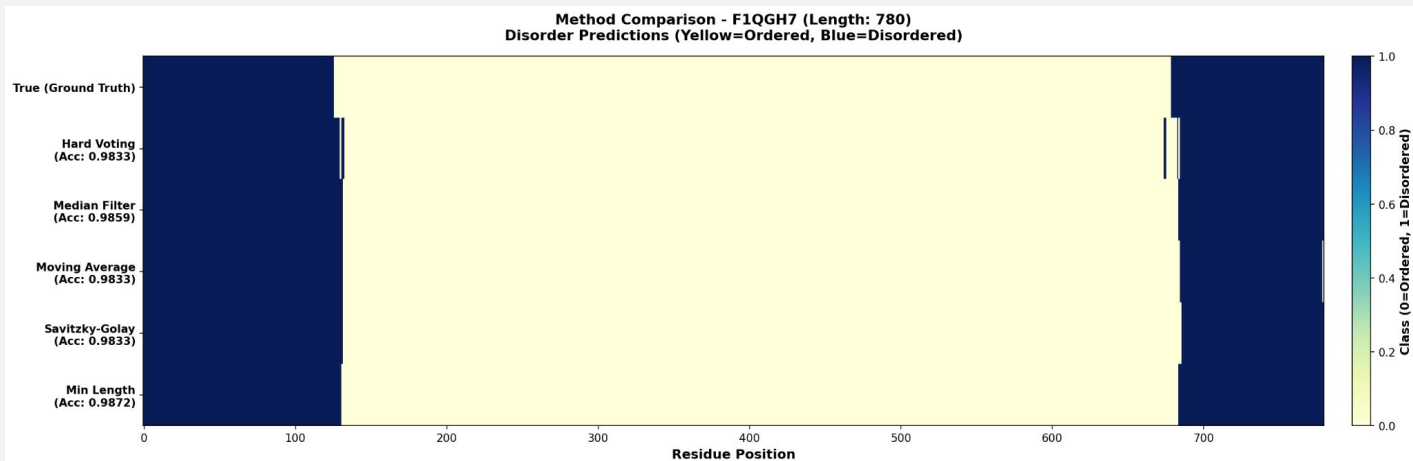
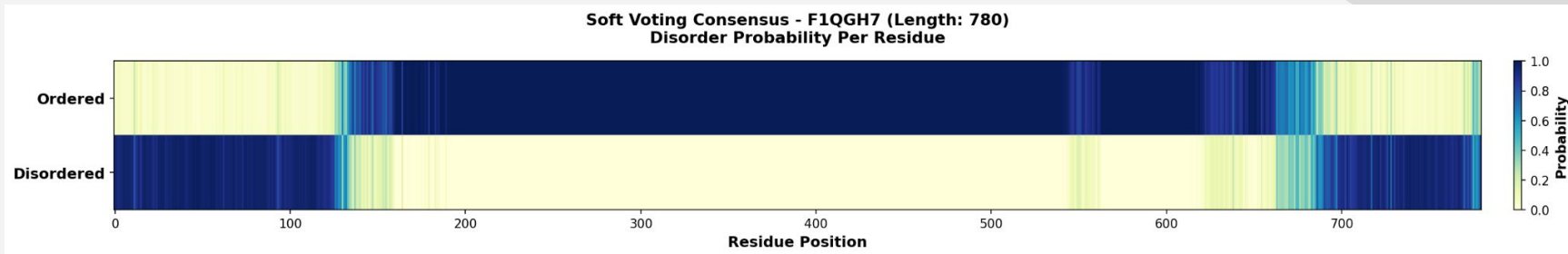


Comparison: Q2KHT3

Protein CLEC16A
(Human)



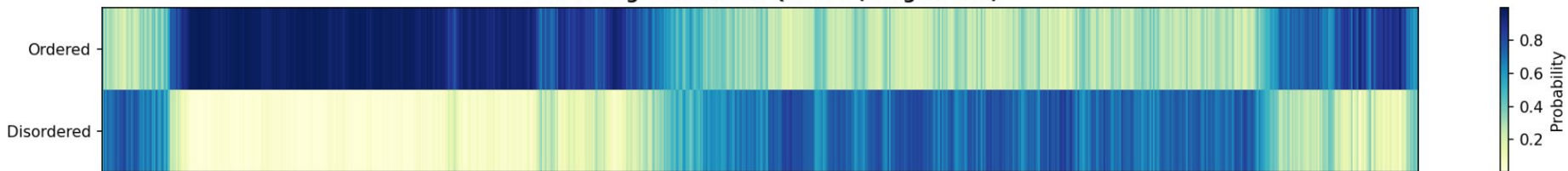
Prediction Example - F1QGH7 Catenin beta-1 using direct sequence-to-disorder state ML model



PDB: 2Z6G

Prediction Example - Q27974 (*Bos Taurus* putative tyrosine-protein phosphatase auxilin) using ESM2-derived model

Soft Voting Consensus: Q27974 (Length: 910)



Prediction Comparison: Q27974 (Length: 910)

